

AUTODRIVE CHALLENGE II™

Year 5 | 2025-2026

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Administrative Procedures

Article A1: AutoDrive Challenge Overview and Objectives

SAE International (SAE) and General Motors (GM) have partnered on the collegiate design competition AutoDrive Challenge™ II. This autonomous vehicle competition series is a five-year competition to develop and demonstrate an autonomous-driving passenger vehicle. The technical goal of the competition is to navigate an urban driving course in an automated driving mode as described by SAE Standard J3016™ level 4 definition by year four, with additional testing and validation in year five.

AutoDrive Challenge II will have collegiate students focus on autonomous mobility technologies and allow for modification and improvement throughout the four-year competition. The focus will be on real-world applications of sensing technologies, computing platforms, software design and implementation, and advanced computation methods such as: image processing, pattern recognition, machine learning, artificial intelligence, sensor fusion, and autonomous vehicle controls.

As part of the SAE International's University Programs portfolio, AutoDrive Challenge II will provide hands-on engineering opportunities for university students to demonstrate innovative solutions to a wide range of challenging opportunities in the rapidly expanding field of engineering systems for automated driving.

Year 1 of the AutoDrive Challenge II will focus on building a solid framework upon which to build systems and solutions for years 2-4. Teams will participate in perception-based challenges, explore the simulation environment, and begin to establish their design for the overall competition.

Year 2 of the AutoDrive Challenge II will focus on vehicle controls and interface development on the Chevy Bolt EUV. Teams will utilize real-time simulators and bench testing in advance of in person testing at Mcity at the University of Michigan in Ann Arbor.

Year 3 of the AutoDrive Challenge II will focus on utilizing software-in-the-loop (SIL) testing to explore edge cases and demonstrate how performance has improved and can continue to keep improving. Teams will use HD mapping to simulate the test environment to complete dynamic missions.

Year 4 of the AutoDrive Challenge II will focus on reaching Level 4 autonomy as defined by SAE standard J3016™, as well as a final Ride Share validation route. In year 4, it is hoped that your team will achieve a minimum of 50% virtual development and validation by utilizing the simulation infrastructure and scenarios that have been created in the past 3 years.

Year 5 of the AutoDrive Challenge II will focus on additional validation testing to demonstrate increased mastery of the skills developed over the previous four years. In year 5, it is hoped that your team will benefit from the opportunity to continue honing critical skills over the extended timeline.

Article A2: Official Competition Announcements & Information

All teams must be familiar with all official announcements concerning the competitions and rule interpretations released by the AutoDrive Challenge II Rules Committee.

AutoDrive Challenge II information and announcements will be provided to students via the following:

- NewsFeed on <http://www.AutoDriveChallenge.com>
- AutoDrive Challenge II App
- Rules updates published online at <http://www.AutoDriveChallenge.com>

Article A3: Competition Judging

CJ.1.1 Good Engineering Practices

Vehicles entered in the AutoDrive Challenge II competition are expected to be designed and built in accordance with good engineering practices and adherence to the AutoDrive Challenge II official ruleset.

CJ.1.2 Judging Categories

The teams will be judged in various static and dynamic vehicle events and reports. Scoring opportunities in technical areas will include Functional Safety Analysis, Engineering Design Concept, Mobility Innovation, MathWorks Simulation Challenge, and Project Leadership. The static events will be scored on appropriate formatting, accurate technological submissions, and adherence to report guidelines. Specific judging criteria for all events within the competition will be laid out in this official rule document.

CJ.1.4 Points Breakdown

Each year of the AutoDrive Challenge II will have 1,000 points possible, and each team will start at 0 points. No points will carry over from previous years.

The below chart summarizes the points breakdown for Year 5:

Events	Possible Points
Concept Design Report & Software Requirements Specification	75
Concept Design Event	75
Project Leadership	75
Mobility Innovation Challenge	75
MathWorks Simulation Challenge	75
Systems Safety & Technical Reports	125
Dynamic Challenges	500
Total Points	1000

CJ.1.5 Competition Awards

AutoDrive Challenge II will present awards to the top 3 teams in each judging category of the event, and to the top 3 cumulative scoring teams for Year 5. Placing teams will receive trophies and monetary awards.

CJ.1.6 Minimum Viable Product Philosophy

The AutoDrive Challenge II encourages student teams to consider how they would approach specific challenges from the perspective of minimum viable product. This philosophical application is designed to facilitate reflection on design processes, and teams are not penalized for designing solutions in validation testing that may not fit the generally understood principles of MVP.

Article A4: Official Competition Language

The official language of the AutoDrive Challenge II Series is English. Document submissions, Rules Questions, presentations, and discussions must be submitted in English.

Article A5: Rules of Conduct

The AutoDrive Challenge II is a technology specific automated vehicle engineering competition which requires performance demonstration of vehicles and autonomous technology integration. Team members and Faculty Advisors will be held to the highest standard of conduct and must abide by their university's own code of conduct, which extends to this competition. It is recognized that this event is an engineering educational experience. In the heat of competition, emotions peak and disputes arise. Our officials are trained volunteers and maximum human effort will be made to settle problems in a reasonable, timely, and professional manner.

RC.1.1 Unsportsmanlike Conduct

In the event of unsportsmanlike conduct, the team will receive a written warning from an official. A second violation will result in expulsion of the team from the competition. Expelled teams are not eligible to receive awards at the conclusion of competition.

RC.1.2 Official Instructions

Failure of a team member to follow an instruction or command directed specifically to that team or team member will result in a twenty-five (25) point penalty.

RC.1.3 Arguments with Officials

Argument with, or disobedience to, any official may result in the team being eliminated from the competition. All members of the team may be immediately escorted from the grounds.

RC.5.4 Problem Resolution

Any problems that arise during the competition will be resolved through the Organizing Committee and the decision will be final.

RC.5.5 Onsite Meetings – Attendance Required

All team members for an event are required to attend the pre-event meeting(s).

RC.5.6 Alcohol and Illegal Material

Alcohol, illegal drugs, weapons, or other illegal material are prohibited on the event site during the competition. This rule will be in effect during the entire competition. Any violation of this rule by a team member will cause the expulsion of the entire team. This applies to both team members and faculty

advisors. Any use of drugs, or the use of alcohol by an underage individual, will be reported to the local authorities.

RC.5.7 Parties

It is expected that no disruptive parties will be held during competition events or workshops and will be prevented by the Faculty Advisor.

RC.5.8 Trash Clean-up

Cleanup of trash and debris is the responsibility of the teams. The team's work area should be kept uncluttered. At the end of the day, each team must clean all debris from their area and help with maintaining an overall clean event site

RC.5.9 Spirit of the Competition

The AutoDrive Challenge II is intended to be a new competition independent from the inaugural AutoDrive Challenge I competition. Teams are expected to utilize the current suite of products, technological solutions, software, and services being provided. Teams who participated in AutoDrive Challenge I are prohibited from recycling software and equipment from AutoDrive Challenge I.

RC.5.10 Onsite Photography

Photography during scored runs of the onsite competition is strictly prohibited. The organizing committee will have a professional photographer & videographer onsite, and the footage and images captured will be made available to teams following the competition.

Article A6: AutoDrive Challenge II Organizing Committee Authority

The AutoDrive Challenge™ II Rules are the responsibility of and owned solely by the Organizing Committee and are issued under the authority of SAE International and General Motors Company. Official announcements from the Organizing Committee, Rules Committee, SAE International and General Motors Company shall be considered part of and shall have the same validity as this rule set. Ambiguities or questions concerning the meaning or intent of these rules will be resolved by the Organizing Committee, SAE International and General Motors Company as appropriate.

OC.1.1 Rules Compliance

By accepting entrance into the AutoDrive Challenge II Competition, the team, members of the team as individuals, faculty advisors and other personnel of the entering university agree to comply with, and be bound by, this rule set and all rule interpretations or procedures issued or announced by the Organizing Committee, Rules Committee, SAE International and General Motors.

Any rules or regulations pertaining to the use of the competition site by teams or individuals, and which are posted, announced and/or otherwise publicly available are incorporated into the AutoDrive Challenge II Rules Set by reference. As examples, all event site waiver requirements, speed limits, parking and facility use rules apply to participants.

All team members, faculty advisors and other university representatives are required to cooperate with, and follow all instructions from, competition organizers, officials, and judges.

The AutoDrive Challenge II rules are to be treated as intellectual property and NOT to be distributed, posted, or made public to anyone not using them for official competition use.

OC.1.2 Understanding the Rules

Teams, team members as individuals, and faculty advisors are responsible for reading and understanding the rules in effect for the competition.

OC.1.3 Participating in the Competition

Teams, team members as individuals, faculty advisors and other representatives of a registered university who are present on-site at a competition are “participating in the competition” from the time they arrive at the event site until they depart the site at the conclusion of the competition.

OC.1.4 Violations on Intent

The violation of intent of a rule will be considered a violation of the rule itself and teams will be penalized accordingly. Questions about the intent or meaning of a rule may be addressed to the Rules Committee through formal channels.

OC.1.5 General Authority

The Organizing Committee, General Motors, and SAE International reserve the right to revise the schedule of any competition and/or interpret or modify the competition rules at any time and in any manner that is, in their sole judgment, required for the efficient operation of AutoDrive Challenge II.

Article A7: Graduate Assistant Funding

AutoDrive Challenge II teams will receive Graduate Assistant funding in August for the upcoming academic year.

Article A8: Travel Allowances, Vehicle Shipping to Event(s), and Key Dates

TA.1.1 Travel Allowance(s)

AutoDrive Challenge II Teams will be given a travel allowance for competitions over the course of the four-year cycle. These amounts will be determined by the Organizing Committee based on location of the workshop and/or event. These funds are to be used for lodging (hotels), meals, and transportation cost of team members only. Teams will typically be notified a month prior to travel of their allotted amount.

For workshops the Organizing Committee will allot a travel stipend to assist in covering costs for all university attendees. This stipend will be predetermined and will be disbursed following the workshop. Attendees are required to validate their attendance by completing the registration survey in advance of the workshop. Teams who have students or faculty who attend a workshop without completing the registration will only be reimbursed for the attendees who completed the survey.

TA.1.2 Vehicle Shipping

AutoDrive Challenge II Teams will also be aided in shipping their competition vehicle. The Organizing Committee will provide shipping from the university to the competition and back to the university once the event has ended. The Organizing Committee will advise on the shipper being used, schedule, and procedure(s).

Teams are asked to designate one team member as the point of contact for all shipping items. All teams, without regard to geographic location of both their university and the competition site, will be required to ship their vehicle to the competition. Teams should plan on the vehicle shipper arriving to take possession of the vehicle 5-7 days before the beginning of competition.

TA.1.3 Key Dates

The Organizing Committee has identified the following dates for official AutoDrive Challenge II events. Teams are expected to attend all events relevant to the competition. Please plan accordingly.

TA.1.3.1 Fall KickOff Event

The AutoDrive Challenge II Year 5 Fall KickOff will be held on September 16th and 17th in Ann Arbor, MI. Please review the handbook for relevant information regarding this event. The Organizing Committee anticipates holding a safety driver training at the Milford Proving Grounds on September 18th, 2025, which is required for anyone who intends to drive the donation vehicle at our onsite event in June. Drivers who have previously completed the training are **not** required to complete the training again.

TA.1.3.2 Winter Workshop Event

The AutoDrive Challenge II Winter Workshop will be held February 2026 in Ann Arbor, MI. Please review the handbook for relevant information regarding this event. We will be conducting the MathWorks Simulation Challenge and the Mobility Innovation Challenge at this event.

TA.1.3.3 Validation Event

The AutoDrive Challenge II Year 5 Onsite Competition will be held May 31st-June 6th, 2026, at Mcity in Ann Arbor, MI. Please review the handbook for relevant information regarding this event.

Article A9 University Continued Commitment

UCC.1.1 Letter of Support

AutoDrive Challenge II Teams must submit a formal letter of support signed by an academic administrator each year of the competition. The signatory should be a dean or other high-level academic administrator. They must include their title in the letter's signature line. They must acknowledge that they are agreeing to meet the commitment from the university to the team/faculty advisor/project in the following areas:

1. Acknowledgement of continued commitment for the full four-year competition.
2. Ability to sign Nondisclosure Agreements and necessary legal support to review and approve any necessary competition documentation.
3. Ability to provide funds above what GM and SAE International have designated for the AutoDrive Challenge II (i.e., additional funding for the full time GRA above the donated funds if necessary).
4. Acknowledgment that the University assumes the liability associated with student project(s).

Letters must be signed and submitted by **November 1** annually.

UCC.1.2 University Site Visit

The AutoDrive Challenge II Program Manager or other designee will conduct site visits to each university to meet with teams, administration, and advisors. The visits will ensure the team is making progress and receiving the necessary support from their university to be successful in the competition. Prior to the site visit teams will be asked to complete a form which aids in preparing both the institution and the visitor.

Article A10: Withdrawal or Expulsion of Team

WE.1.1 Withdrawal

Accepted AutoDrive Challenge II teams who withdraw from the competition must submit in writing a formal letter to the Organizing Committee detailing their decision and must provide signatures from both an academic administrator and the official team faculty advisor.

Withdrawals are final, teams will NOT be permitted to re-enter in any other year. The letter must state that the team wishes to formally withdraw from AutoDrive Challenge II and the reason(s) for the decision.

Teams that withdraw must return any unused portion of the startup funding and graduate assistant funding as soon as possible after the withdrawal letter is accepted by the Organizing Committee and the team is formally withdrawn from the competition. The competition vehicle must be returned or destroyed as directed by General Motors Asset Recovery policies. All component parts and subsequent information received by the team for the AutoDrive Challenge II competition must also be returned.

WE.1.2 Expulsion of Team from Competition

All teams accepted into the AutoDrive Challenge™ II will be held to minimum standards of performance over the four-year competition cycle to maintain their position as accepted into the competition. Teams who do not meet the following requirements may be expelled from the competition:

1. Missing deadlines of written or technical document submissions.
2. Documents are not well presented grammatically or in formatting, or technically accurate displaying sound engineering practices.
3. Failure of teams/universities to be present at key competition events including workshops or the onsite event.
4. Failure of teams to attend onsite technical inspection or any presentations.
5. Failure of teams to comply with their mentor's onsite visit.

The Organizing Committee will provide the team with one written warning and the team will provide a performance plan (in the time specified in the written warning), working with their mentor to increase their level of performance. Failure to acknowledge and comply with this warning will result in immediate expulsion. After such a time as the team has complied with the recommendations they will be returned to non-probationary status. Probationary periods will be determined by the Organizing Committee and will accompany their recommendations.

Teams that are expelled must return any unused portion of the startup funding and graduate assistant funding as soon as possible after the formal expulsion letter is sent by the Organizing Committee and the team is formally expelled from the competition. The competition vehicle must be returned or destroyed as directed by General Motors Asset Recovery policies. All component parts and subsequent information received by the team for the AutoDrive Challenge™ II competition must also be returned.

Article A11: Intellectual Property Creation by Student(s) & University Teams

Intellectual Property created by the students and/or the university team will be initially owned by the student and/or team who thought of the concept or idea that is the subject of the intellectual property. General Motors Company and SAE International as the leading partners in the AutoDrive Challenge II will be permitted access to its usage as it was created during the competition. Any intellectual property created that violates any sponsor or competition non-disclosure agreement cannot be published or presented without written consent of the party owning the intellectual property or copyrighted material and their written legal approval.

All participants will (as per the Non-Disclosure Agreement) agree that ownership of any intellectual property generated by a Participant who is a student and developed during work on the Project is owned by the student and further agree that GM and its affiliates have the right to make, have made, or otherwise use that intellectual property in GM's business activities.

Article A12: Publishing

Students and/or university teams are permitted to publish materials that they have developed or researched throughout the AutoDrive Challenge II. Any material that is used in the published work and has/refers to any sponsor companies copyrighted material, or intellectual property must be approved by the company from which it was from. Please use the Publication Request Form in Series Resources on www.AutoDrivechallenge.com.

Participants are encouraged to read: "How to Write an SAE International Technical Paper" - http://volunteers.sae.org/authors/sae_tech_paper.pdf. Page 4 will help in citing references.

Article A13: Sponsorship and Donated Products

AutoDrive Challenge™ II will have Official Supplier Partners and Integrated Suppliers who will donate to the competition with technical support and products. As these sponsors are confirmed we will communicate the specifics of their donations, support, how to contact them, and the delivery/disposal/return of products/services. We will publish this information on both the <http://www.AutoDriveChallenge.com> site and <https://www.sae.org/attend/student-events/AutoDrive-challenge/sponsor?tab=2>.

Article A14: Mcity Facility Rental

The AutoDrive Challenge™ II competition will be held annually at the Mcity Testing Facility in Ann Arbor, Michigan. The Organizing Committee recognizes this is a public facility and is available for testing rental, however, in the spirit of fair competition teams participating in the AutoDrive Challenge II are prohibited from reserving the Mcity Testing Facility for testing beginning August 1, 2021, and ending July 31, 2026. **This restriction also applies to sponsors for individual teams providing data from their own testing at Mcity or capturing data at the Mcity Testing Facility on behalf of teams.** If your team has any questions, please submit them on <https://www.AutoDrivechallenge.com>.

Participant Requirements & Registration

Article B1 : Participant Online Registration

AutoDrive Challenge™ II teams selected by the Organizing Committee based upon submitted proposal acceptance will be listed on the Registered Teams List and students can then access their online team profile page on the sae.org website.

Article B2: Individual Team Member Requirements

TMR.1.1 Eligibility Limits

Eligibility is limited to full-time undergraduate and graduate students to ensure that this is a collegiate engineering design competition. Part-time, degree seeking undergraduate students may be considered eligible to participate in AutoDrive Challenge II on a case-by-case basis.

TMR.1.2 Student Status

Team members must be enrolled as degree seeking undergraduate or graduate student(s) in the college or university of the team with which they are participating. Teams cannot be formed with members from two or more universities.

TMR.1.3 Society Membership

Team members must be members of SAE International. Students can join SAE online at: www.sae.org/students Students should select the “Student Membership” link and follow the series of the questions that are asked. Faculty who wish to be SAE members should choose the “Professional Membership” link and proceed to the series of questions.

Please note all student participants must be SAE International members to participate the AutoDrive Challenge II. It is not mandatory for faculty to join but they must be on the team list on sae.org.

TMR.1.4 Age

Team members must be at least eighteen (18) years of age, at the time of participation on the competition team.

TMR.1.5 Driver's License

Team members who will be a safety driver in the competition vehicle at any time during a competition must hold a valid, government issued driver's license and will be asked to present it onsite at Technical Inspection.

TMR.1.6 Liability Waiver

All on-site participants, including students, faculty, and volunteers are required to sign a liability waiver upon registering onsite.

TMR.1.7 Medical Insurance

Individual medical insurance coverage is required and is the sole responsibility of the participant.

TMR.1.8 Online registration

Every participant, including advisors must affiliate themselves and complete the following information on under the team's registration page on the SAE website
<http://students.sae.org/cds/AutoDrive/event/registration/>

- Emergency contact data (point of contact (parent/guardian, spouse, or other appropriate designee), relationship, and phone number)
- Participants (your) cell phone number
- Shirt size
- Dietary restrictions
- Accessibility Needs
- Medical Conditions
- Country of Citizenship

TMR.1.9 Onsite Registration Requirement

Onsite registration will be required of all team members, graduate assistants, and faculty advisors. Official registration procedures will be addressed in the student handbook and on the website for teams. This information will be published in advance of onsite competition.

Article B3: Faculty Advisor Requirements

Each team is expected to have a dedicated AutoDrive Challenge II Faculty Advisor appointed by the university. The Faculty Advisor is expected to accompany the team to the competition as well as all appropriate workshops and trainings and will be considered by competition officials to be the official university representative.

1. Faculty Advisor(s) may advise their teams on general engineering and engineering project management theory.
2. Faculty Advisor(s) may not design any part of the vehicle including any part of the autonomous computing/driving system(s) nor directly participate in the development of any documentation or presentation.
3. Additionally, Faculty Advisor(s) may not fabricate nor assemble any components nor assist in the preparation, maintenance, testing, or operation of the vehicle.

In Brief – Faculty Advisor(s) may not design (including software and calibration), build, or repair any part of the car, including any part of the autonomous computing/driving system(s).

Article B4: SAE Technical Standards Access for Participants

A cooperative program of SAE International's Education Board and Technical Standards Board is making some of SAE International's Technical Standards available to teams registered for any North American Collegiate Design Series (CDS) or Emerging Technology Series (ETS) competition at no cost. The Technical Standards referenced in the CDS & ETS rules, along with other standards with reference value, will be accessible online to registered teams, team members and faculty advisors.

To access the standards:

- 1-your team must be registered for AutoDrive Challenge II
- 2-the individual team member or faculty advisor wanting access must be affiliated to the team on SAE International's website (www.sae.org).

In 2016, the SAE Digital Library became SAE Mobilus.

- Navigate to your profile.
- Click on your Team Name.

- Click the SAE Mobilus link.

Search standards either by J-number assigned or topic of interest such as brake light. A list of accessible SAE Technical Standards can be found in Appendix.

Article B5: Non-Disclosure Agreements

NDA.1.1 General Motors Non-Disclosure Agreements

All AutoDrive Challenge II Team Members, Team Captains, Graduate Assistants and Faculty Advisors will be required to sign non-disclosure agreements due to the sensitive nature of intellectual property that is given to teams with the donated GM Chevrolet Bolt EUV. As more team members are added they will not be considered official members of the team until they upload their signed NDA onto <http://www.AutoDriveChallenge.com>

NDA.1.2 Sponsors Non-Disclosure Agreements

All AutoDrive Challenge II Team Members, Team Captains, Graduate Assistants and Faculty Advisors will be required to sign non-disclosure agreements due to the sensitive nature of intellectual property that is given to teams with the donated components and parts from sponsors. As more team members are added they will not be considered official members of the team until they upload their signed NDA onto <http://www.AutoDriveChallenge.com>. As more sponsors are determined each sponsor may ask for an NDA for their product donation and participation in AutoDrive Challenge II.

Article B6: General Motors Mentor Assignment

GMM.1.1 Assignment

General Motors will provide an experienced engineer to each team to mentor and guide them through the four-year competition. Mentors will also serve as a liaison to teams on behalf of GM.

Mentors will provide progress reports to the competition organizers. Participating universities will receive additional mentoring support from competition sponsors to help the teams use the donated components, software, and other products through workshops.

GMM.1.2 Mentor Roles

Team mentors are to guide teams through the four-year competition but are not a member of the team. Their role is to:

1. Act as a sounding board for teams as they define their concept selection and help the team arrive at sound engineering practice decisions as they navigate the learning process for this new technology and integration keeping them within the scope of the competition and helping the students learn by trial and error.
2. Provide constructive educational feedback on technical reports.
3. Educate and enforce safety practices with the GM donated vehicle.
4. Be familiar with the official competition rules and event operations to help guide the teams in becoming compliant with both.
5. Advocate proper storage, use, and management of GM intellectual property information amongst teams.
6. Communicate with teams bi-weekly (twice a month) to make sure they are on track with the competition timeline, either onsite at their university or a conference/video call.

NOTE: Mentors will not assist in the hands-on integration of systems, writing of official documentation including the team's concept and design presentations, nor will they be allowed to participate as a team member in the onsite team presentations.

GMM.1.3 Reporting

Mentors will document and report their teams' status to the competition organizers following each team interaction.

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Rules/Technical Questions & Document Submission(s) Procedures

Article C1: Account Signup for Online Submission(s)

RQ.1.1 Account Creation

Account creation for online document submission through <http://www.AutoDriveChallenge.com> requires the following steps:

1. Visit <http://www.AutoDriveChallenge.com>
2. Follow the account signup instructions on the website. Select “Your Affiliation” as either “Faculty Advisor” or “Team Captain/Team Member.”
3. Faculty Advisor ONLY: Will use the SAE “authentication number” for AutoDrive Challenge™ II signup. This is the SAE confirmation number that was issued when your team registered on the sae.org website.

NOTE - There may be a delay of up to three (3) business days between the time your team registers for a competition and the AutoDrive Challenge II site recognizes the validity of your authentication number and the faculty advisor may create an account. Once your faculty advisor has created an account it will remain valid for the full four years.

RQ.1.2 Faculty Advisor Role

Each team must have at least one person identified on AutoDrive Challenge II as the Faculty Advisor. The Advisor has unique responsibilities on the site including accepting other team members for site access. Until the faculty accepts a member's signup that person cannot upload or view team documents. Faculty Advisors automatically have the same roles and privileges as their team members.

RQ.1.3 Team Member Restrictions

Team members must be approved by the Faculty Advisor before being able to view or upload team documents. Once a team member leaves the team or graduates, they **MUST** be removed immediately by the Faculty Advisor removing their access to any competition materials/documents etc.

RQ.1.4 Uploading Documents

All team members and the team captain have equal authority to upload and/or replace documents in the name of the team. The last version uploaded when judges access the submissions is what will be scored.

RQ.1.5 Document Access

Team uploaded documents can only be viewed by

1. Members of the submitting team,
2. Authorized judges, technical inspectors, and officials
3. Organizing Committee Members

Reminder – The website does not know what you intended to submit or what you thought you were doing. Anything your team uploads to the site is an official action by your team.

NOTE – All Non-Disclosure Agreements need to be uploaded and approved before team members have access to documents. All team members need to be affiliated on the AutoDrive Challenge II website as the Non-Disclosure Agreements will be present on that site.

Article C2: Official Question & Answer Process

All AutoDrive Challenge II rules or technology (sponsoring company) questions will be asked through the official website <http://www.AutoDrivechallenge.com>.

QA.2.1 How to Submit Rules-Related Questions

1. **Create an AutoDrive Challenge II account**

This same account is used for submitting competition reports and documents, submitting rules questions, etc.

2. **Affiliate to your Team/University**

Not everyone who has an AutoDrive Challenge II account is a member of a student team. If you are, be sure to affiliate your account with the correct School and Team at that school so that you can use the Team Member parts of the AutoDrive Challenge II system.

3. **Login and go to "Ask a Rules Question" under the Team Member menu**

Choose "submit a new rules question". Also, available on this menu is "My Team's Rules Questions", where you can view all the questions your team has asked, and the answers given for those questions. You will also be able to add more information / attachments, respond to questions back from the Rules Reps, and browse published Rules FAQs.

QA.2.2 Response Time

Please allow a minimum of two (2) weeks for a response. The Rules Committee will respond as quickly as possible, however responses to questions presenting new issues, or of unusual complexity, may take more than two weeks. Please do not resend questions.

Teams are expected check the FAQs before posting an inquiry. The Committee will answer questions that are not already answered in the rules or FAQs or that require new or novel rule interpretations. The Committee will not respond to questions that are already answered in the rules.

Article C3: Document Submission(s) & Deadlines

DS.1.1 Required Documents and Required Forms

The following documents must be submitted by the action deadlines posted on the competition website or otherwise published by the organizers.

All forms are located at <http://www.AutoDrivechallenge.com>

DS.1.2 Deadlines

Volunteer judges evaluate all the required submissions, and it is essential for them to have enough time to complete their work. There are no exceptions to the document submission deadlines and late submissions will incur penalties. Please note different documents or submissions have different deadlines.

Carefully read the event rules and check the website of the competition. Teams will **not** be notified if a document is submitted incorrectly. Each team is responsible for confirming their documents have been properly uploaded and that the deadlines have been met. **Email confirmations to teams will not be sent.**

Article C4: Competition Penalties

CP.1.1 Late Submission Penalties

Documents or submissions uploaded after the deadline, or submitted largely incomplete within the deadline, will receive a 10%-point penalty per day based on the total value of the submission, based on the time difference between the deadline and the actual date and time of receipt or upload, subject to official discretion. Submissions are due by 11:59:59 PM Eastern on the date listed above. Submissions received after this will be subject to penalties. Submission times are based on the time the server registers the uploaded document, not at the time the document upload was started.

CP.1.2 Late Document Submissions

Teams are expected to complete documents and to submit them by the articulated deadlines. Teams are required to submit all deliverables as outlined in this rules document. The submission portal will close 5 days after the scheduled due date. **Submission reminders will not be sent to teams.**

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Static Events

Article D1: Concept Design Report

CDR.1.1 Concept Design Report Objective

The purpose of the Concept Design Report is for each team to document and showcase their design for each Year of the competition. Your design is expected to evolve throughout the competition, and the report objectives of the Concept Design Report will reflect these changes annually.

This year marks the culmination of five years of development, innovation, and iteration. For the Year 5 Concept Design Report, your team will provide a comprehensive review of the design evolution and final implementation of your autonomous vehicle. This report should highlight how your design choices have matured over the last five years and contributed to the current state of the vehicle.

We look forward to seeing how your team's long-term efforts and technical decisions have come together in the culmination of this five-year journey.

Your report must cover the following Concept Design Elements:

1. Sensor Suite
2. Sensor Fusion
3. Computer, Thermal, Electrical Design and Implementation
4. Software Architecture
5. Positioning and Path Planning
6. Object Detection and Identification
7. Vehicle Dynamics

For each design element, clearly explain the implementation on your vehicle and rationale behind your decisions. For example, justify your sensor placement positions. Why were those locations chosen over other alternatives?

Please refer to the outline and scoring rubric below for detailed expectations. The Concept Design Report is due on 4/24/2026.

CDR.1.2 Concept Design Report Content & Format

This section details the preferred report outline and specific contents, as well as report formatting.

CDR.1.2.1 Concept Design Report Outline

1. Introduction
 - Include a summary of the competition, the team, and report contents.
2. Sensor Suite
 - Detail your Year 5 sensor suite. Describe placement on the vehicle and why those areas were chosen. Include hardware mounting details. Show test results where feasible to support decisions.
3. Sensor Fusion
 - Explain why a variety of sensors is needed and your fusion methodology. Show test results where feasible to support decisions.
4. Computer, Thermal, Electrical Design and Implementation
 - Detail your compute placement, thermal design, and electrical layout. Explain why these designs are ideal. Show test results where feasible to support decisions.
5. Software Architecture

- Detail the architecture of your software. Explain why this is the ideal configuration.
- 6. Positioning and Path Planning
 - Detail the design and methodologies for vehicle positioning and route/path planning. Show test results where feasible.
- 7. Object Detection and Identification
 - Detail your methods for detecting objects and for identifying them. Show test results to support why these methods are ideal.
- 8. Vehicle Dynamics
 - How do you control your vehicle acceleration, braking and steering. Show test results to support why these methods are ideal.
- 9. Conclusion

The Concept Design Report must not exceed twenty-five (25) pages:

- Ten (10) pages of text
- Five (5) pages of drawings
- Ten (10) optional pages containing content to be defined by the team (photos, graphs, analysis, text, etc.)

All Concept Design Reports should be put into SAE Technical Paper Formatting (http://volunteers.sae.org/authors/sae_tech_paper.pdf). See website for format. In the event of conflicting requirements, this Concept Design Report Article supersedes the SAE Technical Paper Formatting (for example, total report length).

The judges (professional engineers from the AutoDrive Challenge™ sponsor companies) will evaluate the Concept Design Report based on the following rubric.

Intro & Design Elements	Score
Introduction	
Introduction of the competition, your team, and report contents	2
Sensor Suite	
Detail your Year 5 sensor suite. Placement on the vehicle and why those areas were chosen. Hardware mounting details. Show test results where feasible to support.	4
Sensor Fusion	
Detail the fusion of the sensors in your suite. Why a variety of sensors is needed. Your fusion methodology. Show test results where feasible to support.	4
Computer, Thermal, Electrical Design and Implementation	
Detail your compute placement, thermal design, and electrical design. Why are these designs ideal. Show test results where feasible to support.	4
Software Architecture	
Detail the architecture of your software. Why is this the ideal configuration.	4
Positioning and Path Planning	
Detail the design and methodologies for vehicle positioning and route/path planning. Show test results where feasible.	4
Object Detection and Identification	
Detail your methods for detecting objects and for identifying them. Show test results to support why these methods are ideal.	4
Vehicle Dynamics	
How do you control your vehicle acceleration, braking and steering. Show test results to support why these methods are ideal.	4
Total Score	30
Format and Documentation	Score
Report Format	
Report does not exceed 25 pages	1
Grammar is correct	1
SAE Technical Paper Formatting is followed	1
Documentation	
List of all papers written, conferences attended, presentations held, and patents developed throughout the competition present	1
List of references present	1
Total Score	5
Total Overall Score:	35

Article D2: Concept Design Event

CDE.1.1 Event Objective

Based on Challenge I Team feedback, the Concept Design Events will be open for all teams to view. This event provides insight and learnings for each team to learn and grow from one another. The topics for the competition may change each year, but the event will always be held during the competition.

Viewer excitement and engagement are paramount for this event, and as such, the presentation should be:

- Personal (what did your team do?)
- Concise (you get 20 minutes)
- Engaging (presentation skills and practice matter!)

Take us on a journey, highlight key decisions, setbacks, or insights your team encountered.

For the Year 5 Concept Design Event, each team must choose two of the four provided Concept Design Report Elements as shown below. Please reference Article D1 for descriptions. For each element:

- Detail the current implementation on your vehicle
- Explain why this design was chosen, and
- Describe how this design evolved from Year 1 to Year 5.

Additionally, include one of the following in your presentation:

- Your biggest lesson learned, **or**
- Your team's competition strategy throughout the 5 years

Concept Design Presentation Part 1:

Please choose **two** Concept Design Elements below to review in your presentation.

1. Sensor Suite
2. Computer, Thermal, Electrical Design and Implementation
3. Positioning and Path Planning
4. Object Detection and Identification

These are the same topics covered in your Concept Design Report (see Article D1).

Concept Design Presentation Part 2:

Please choose **one** of the options below to review in your presentation.

1. Biggest Lesson Learned
 - (a) Describe the most impactful lesson your team learned during competition.
 - (b) What caused it? How did you respond and improve as a result?
2. Competition Strategy
 - (a) What strategies, tips, and tricks did your team use to gain points within competition constraints?
 - (b) What worked well, what didn't?

To support your success, each team will be scheduled for a 15-minute check-in at the Winter Workshop to answer questions and review your plans for presenting in June. Teams are also welcome to get feedback on their SRS deliverables at this time.

CDE.1.2 Judging Procedures - Onsite

- Teams will have up to 20 Minutes to present. There will be an additional 5 Minutes of Q&A from judges after the presentation.
- Your **vehicle will NOT BE present** at the presentations. In lieu of your vehicle being present at the presentations, judges will visit your paddock separately at an unscheduled time during competition hours (see CDE.1.4 for more details).
- The presentation itself will not be interrupted by questions.

Presentation Guidelines

- Teams can select up to 6 student members. We highly recommend at least half of the presenters be undergraduate students. All presenters should add value to the presentation.
- Faculty and the GM Mentor are not permitted to present during the Design Event or answer questions for the team.
- Up to 10 judges will score each presentation. These judges will be volunteers from sponsor companies and be from various backgrounds in the Autonomous Vehicle industry.

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CDE.1.3 Concept Design Event Judging Rubric

The judges (professional engineers from the AutoDrive Challenge II sponsor companies) will evaluate the Concept Design Event based on the rubric below.

Concept Design Event	Score
Presentation Style	
Was the presentation interesting and engaging? Did it tell a story?	25
Presentation Content	
Part 1: Concept Design Elements	
Teams review two of the Concept Elements and the implementation currently on their vehicle:	
1. Sensor Suite 2. Computer, Thermal, Electrical Design and Implementation 3. Positioning and Path Planning 4. Object Detection and Identification	10
Discuss details on why this design was chosen. Data is shown (Pugh Matrix, Plots, KT Matrix, TRIZ, test results, or otherwise) to prove why the implemented solution was the best decision.	20
For the two concept elements, detail how they have changed from Year 1 to Year 5 and why.	10
Part 2: Biggest Lesson Learned OR Competition Strategy	
Teams review one of the items below:	
1) Biggest Lesson Learned a) Was this an impactful lesson? Did the team detail what happened, what they learned, and how they pivoted?	
2) Competition Strategy a) Did the team provide a clear and impactful strategy? Did they explain what worked well and, if applicable, what did not work well with the strategy?	10
Total Overall Score:	75

CDE.1.3 Judging Process

Competition judges will observe and score the quality of student teams' responses to the Report and Event challenges. Judges will evaluate the performance of the participants per the standards and criteria provided.

All decisions by judges are final and not subject to critique, challenge, or reconsideration under any circumstances. Participants are not allowed to approach judges to solicit feedback or comments outside of what is provided by the competition process under any circumstances. Violation of this rule could result in being disqualified from the competition.

A judge may abstain from input, voting on a team, or scoring if they feel that their participation in the round of competition reflects a conflict of interest due to prior relationship (e.g., employment, academic, familial) with individual team's members or if participation will create an appearance of impropriety.

CDE.1.4 Paddock Judging

New for Year 5, Concept Design Judges will be visiting your team's paddock area to:

- View the vehicle
- Speak with team members, and
- Ask general questions about the system and design

No scheduled time will be provided, teams must be prepared throughout the event.

One or more judges may drop by at any time to visit your paddock when your vehicle is present. You will need to ensure that all members of your team are knowledgeable about the general design of your vehicle. Judges will be viewing your vehicle, paddock space, and asking questions relating to the overall design of your vehicle. Please review the rubric below to best prepare for this activity.

Paddock Judging Informal Event	Score
Tidiness Was the paddock space tidy and presentable? Free of trash?	5
Design Tour - Was a team member able to walk you around the vehicle and provide an overall design tour of major components (sensor suite, compute rack, HMI)? - Was the team member able to answer reasonable questions about the overall design of the vehicle, respectfully admit that the question is outside of their area of expertise, or defer to another team member at the paddock to help better answer the question?	5
Professionalism and Courtesy Were all team members that you spoke with courteous and professional during your interactions with them?	5
Total Overall Score:	15

Article D3: Software Requirements Specification (SRS)

SRS.1.1 SRS Objective

Each team is required to maintain and submit a Software Requirements Specification (SRS). The intent is to document your system and software design requirements based on competition rules. These requirements will serve as a guiding document for your team, for the design of your vehicle, and for use in testing when implementation is complete for each area.

The Software Requirements Specification Draft is due November 30th (5 points), and Final is due May 1st (45 points). For Year 5, the same deadlines for the deliverables will apply. Continuing from Year 4, teams are to test their requirements and include their test results as part of the SRS Final deliverable.

The Draft must demonstrate progress from the previous year's Final submission and include an outline of how each requirement will be tested, a summary of test status (if applicable), and an appendix for test results. If development in an area is complete, teams are encouraged to submit actual test results with their Draft. Judges will provide feedback on the Draft, and teams should incorporate this feedback into their Final submission.

The Final submission must include any updated or additional requirements, as well as the complete test results. The test results should detail the test platform, pass/fail metrics, and comments related to the test procedure. Comments must be provided for each test, explaining whether it passed or failed, and should address whether it met or exceeded expectations, and by how much. Please see SRS 1.3 for more details regarding test results for Appendix 1.

SRS.1.2 SRS Content & Format

To assist judges in evaluating your SRS, please follow the outline provided. While the format may be customized for your team, the structure should remain consistent. Examples have been provided for certain sections to clarify expectations.

Requirements should reflect the current year's rules and consider industry trends to develop requirements based on existing knowledge and best practices.

The SRS document should not exceed 30 pages to allow judges to evaluate all teams in a timely manner. Do not submit multiple files. Appendices for test results and open-source software disclosures must be included within the SRS document.

SRS.1.3 Software Requirements Specification Outline

1) Introduction

- a) Describe the document's purpose, intended audience, and scope and usage of the requirements.
- b) Define the system functionality covered (vehicle CAN, sensors, input/output scope)

2) Assumptions and Dependencies

- a) Identify assumptions or dependencies that influence your requirements.
- b) *Example: "Our team assumes no rain or snow during vehicle operation."*

3) System Features and Requirements

a) Functional Requirements

i) Vehicle Dynamics

- (1) Include requirements for CAN interface: braking, acceleration, steering, etc.
- (2) *Example: "The vehicle shall never travel faster than x kph."*

ii) Sign & Lane Detection

- (1) Example: "The system shall detect stop lines when the vehicle is within x meters away."
- (2) Example: "The system shall detect green lights within x meters 95% of the time."

iii) Object Detection & Tracking

- (1) Road signs, pedestrians, vehicles, animals, etc.
- (2) Example: "The system shall detect any moving objects within x meters of the vehicle and calculate if/when it will collide with the vehicle."

iv) Positioning & Route Planning

- (1) Example: "Given an address, the system shall calculate a route to the destination based on the vehicle's current location within x milliseconds."
- (2) Example: "The accuracy of the vehicle's position shall be within x centimeters."
- (3) Example: "The vehicle shall initiate a lane change if the planned route has a blockage for more than x seconds."

v) AV Mode Blue Light

- (1) Example: "Upon entering Autonomous Mode, the system shall illuminate the external and internal AV Mode Light via the hardware interface within x milliseconds at a frequency of y hertz."

vi) V2X

- (1) Include requirements around V2X (Vehicle to Infrastructure / Vehicle to Vehicle) integration.
- (2) Example: "Vehicle shall respond to speed limit information received within x milliseconds."

b) Non-Competition Requirements

- i) Use this section to include any self-imposed vehicle/cart level requirements that the team has decided to include beyond the requirements transposed from the rules.
- ii) Example: "The vehicle shall have a top speed of 15mph."
 - (1) Rules state 25mph, but team may feel their vehicle should travel slower than this.
- iii) Example: "The vehicle shall play a chime when the user has arrived at their destination."

c) External Interface Requirements

- i) For interfaces coming into our system, what are these signals and what are the expected ranges?
- ii) Example: "The system shall receive x signal at a frequency of y hertz."

4) Conclusion

5) Appendix 1: Test Results

For each requirement, provide test results in table format. Each requirement in the report must have an associated test line item in this section.

The table is required to include the following columns:

- a) SRS Requirement Identifier – Refer to the traceable requirement ID and title. Teams do not need to restate the requirement description. *(Draft and Final Required)*
- b) Test Platform – Was the test performed on the vehicle, simulation, test bench, or multiple? *(Draft and Final Required)*
- c) Pass / Fail / Unable to Test *(Final Required)*
- d) Comments with respect to the testing procedure and result– Briefly describe how you tested the requirement. Comments must be provided for each test, explaining whether it passed or

failed, and should address whether it met or exceeded expectations, and by how much. (*Final Required*)

- Did the requirement need to be updated?
- If the test failed, is it acceptable to proceed? Why?
- If you were unable to test the requirement, why?
- If you tested the requirement with multiple platforms, how did they compare?

Include a summary paragraph describing your Year 5 testing process:

- What surprises did you encounter?
- Did any requirements need updates to be testable?
- Were any requirements added or changed as the vehicle developed and changed?
- What were the most difficult requirements to verify?
- What tools, strategies, or methods were used to validate?

6) Appendix 2: Open-Source Software Disclosure Form

Attach your team's Open-Source Software Disclosure Form using the provided template. Include:

- All software packages and bundles leveraged that you did not develop
- Packages used directly (e.g. ROS, dSPACE)
- Operating systems
- Any third-party or external code that was not developed by your team

Ensure that your submission follows the OSS disclosure template Excel sheet and that all columns appear on one page.

SRS.1.4 Software Requirements Specification Judging Rubric

The judges (professional engineers from the AutoDrive Challenge™ sponsor companies) will evaluate the Software Requirements Specifications based on the rubric below.	Score
Introduction & Assumptions/Dependencies	1
Functional Requirements (broken down by section below) Minimal points for having requirement(s) present Half points for partially developed requirements Full points for completely developed requirements	
Vehicle Dynamics	1
Sign & Lane Detection	1
Object Detection & Tracking	1
Positioning & Route Planning	1
AV Mode Blue Light	1
V2X	1
Non-Competition Requirements Full points for fully developed requirements	1
External Interface Requirements Full points for fully developed requirements	1
Appendix 1 – Test Results Test results follow the format described in the rules including all columns and all requirements from SRS	15
Summary concisely describes the team's testing during Year 5 and addresses all questions posed in the rules	5
Appendix 2 – Open-Source Software Disclosure Form Complete the Opensource Software Disclosure Form using the provided template	1
Total Score	30
Format and Documentation	Score
Report Format Document follows the outline set forth in the rules	5
Grammar, easy to follow the format that is chosen by team (no set format was given in the rules)	
Total Score	5

Article D4: Mobility Innovation Series (MIS)

AutoDrive Challenge II introduces the Mobility Innovation Series (MIS) to participating teams. This series of challenges will require teams to develop innovative, multi-disciplinary solutions to real-world problems. The goal of these challenges is to introduce the many facets of autonomous & electrified vehicle development, as well as supplement each team's individual vehicle development cycles. These challenges may facilitate inclusion of many non-engineering focused students, such as those in economics, marketing, social media, and management, and still require engineering input.

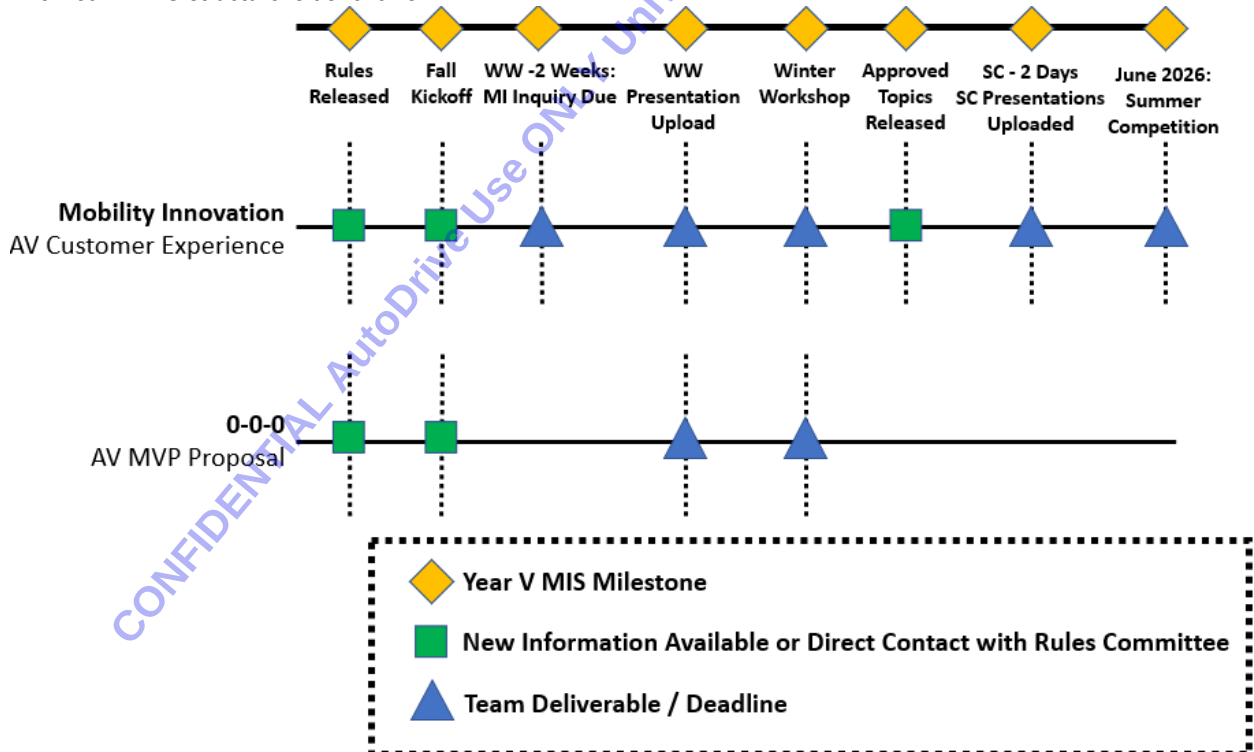
MIS.1.1 Mobility Innovation and 0-0-0 Challenges

Year V will have one Mobility Innovation challenge and one Zero-Zero-Zero (0-0-0) challenge. Innovation challenges will focus on a variety of aspects, ranging from vehicle-to-vehicle interaction, vehicles to the environment, data collection, cybersecurity, and others. 0-0-0 challenges, on the other hand, will align with on General Motor's mission of Zero Crashes, Zero Emissions, and Zero Congestion by 2030.

MIS.1.2 Structure and Timing

Year V's Mobility Innovation Series will follow a similar structure to year IV. Teams are expected to participate in each challenge in addition to other deliverables as outlined in the AutoDrive Challenge II Rules.

The Year V MIS structure is as follows:



Timing overviews will be provided within each Challenge's ruleset.

MIS.1.3 Deliverables and Scoring

Each Mobility Innovation challenge will have several deliverables which vary per challenge. Year V MIS Challenges will total 75 points: 45 points for the Innovation Challenge and 30 for the 0-0-0 Challenge. See each individual challenge's overview for scoring details.

MIS.1.4 Violation of Rules

In addition to the rules outlined for these challenges, the rules of ethics and academic integrity apply, meaning that plagiarism, using faculty in the writing of papers or presentations, and theft of other team's ideas will be considered a violation and the team eliminated from the competition. All sources for information, data, research, and reports must be cited using MLA or AMA style of writing, format and citations. Any violation of these rules either intentionally or unintentionally will result in an appropriate sanction that includes assessment of a penalty, disqualification from the competition and/or the return of prize money. The imposition of sanctions and penalties is within the sole discretion of the Mobility Innovation Rules Committee ("MIRC") and is not subject to discussion, debate, or challenge.

Article D5: Mobility Innovation Event (MIE) Year V Innovation and 0-0-0

MIE 1 Innovation: AV Customer Experience

MIE.1.1 Problem Statement

OEMs are always looking for opportunities to increase their market share, whether through the introduction of attractive new vehicles or the incorporation of advanced features. As autonomous vehicles evolve, OEMs need to ensure that customers are not only open to AVs but also enthusiastic about using them, particularly since some may be reluctant to transition from traditional driver-operated vehicles. A key strategy for promoting this adoption is to improve the customer experience and set themselves apart from competitors.

MIE.1.2 Challenge & Considerations

Teams will take the position of a start-up focused on improving the AV experience for customers. Your team is challenged to present a product pitch at Summer Competition to industry professionals as if you intend to gather seed funding. To consider investing in your product, potential investors need to understand your team's motivation, the details of the product, and your business plan. Your presentation should open with an engaging elevator pitch and close with a strong appeal for support. Teams may target Ride Share, Personal, or both types of AVs.

Notes:

- The proposed product/feature must be present on the AV, either physically or through the AV's internal software. For example, reward-based app/systems designed to encourage AV use will not be permitted.
- Teams must propose a novel product to improve customer experience. In other words, the solutions should not already be commonly utilized or self-evident within the AV space.

MIE. 1.3 Challenge Definitions

- OEM: Original Equipment Manufacturer. The final assembler of the complete vehicle, whether it is a passenger, commercial, or off-road vehicle.
- AV Customer Experience: Encompasses the overall journey and interactions a customer has with an AV from vehicle entrance to exit. Customer experience can be characterized by many factors including ease of use, satisfaction, and emotional connection with the brand.
- KPI: Key Performance Indicator. A measurable value that demonstrates how an organization achieves self-determined key objectives.

MIE 1.4 Deliverables

Teams shall be presenting their solution proposal to a live audience at the Summer Competition. Teams will be scheduled in blocks of 30 minutes. Teams are permitted to present for a maximum of 15 minutes, with an additional 10 for Q&A from the judges and audience.

Presentation content shall include and address, at minimum, the following:

- Engage the audience: Start with a 60 second elevator pitch that captures the judge's attention.
 - Series resources and reference guides will be provided at Fall Kickoff.
- Define the problem statement: Identify and justify a product or feature that improves the customer's AV experience. This can solve an existing problem or enhance one's experience.
 - Justification should include background research and measurable data that establishes a problem to be solved.
- Pitch the solution: Sell the audience on your idea. Pitches should address the following:
 - Stakeholders & Impacts
 - Who are the primary beneficiaries?
 - How does the product or feature help enhance their experience?
 - Benchmarking & Market Assessment
 - Has anyone else tried to address the problem?
 - If not, why hasn't the problem been addressed yet? What challenges currently exist?
 - Team's Solution: Technical & Customer-facing Details
 - What is the team's solution?
 - What differentiates your product from others?
 - Share a system diagram and/or CAD/simulation that explains how the product works.
 - How does it integrate into the vehicle?
 - Explain how the customer would experience the product / feature.
 - Business Plan
 - What is your business plan?
 - How does this make money, draw more customers, or meet other self-defined KPIs?
 - How much does the solution cost to implement? This should include estimates for hardware, software, and engineering design & development.
- Leave a strong impression:
 - Concluding statements should outline final thoughts and end with a strong appeal for funding and support.

MIE 1.4 Timing and Enablers

The Mobility Innovation Rules Committee has set forth the following as a high-level project management plan. Note that there are deliverables that all teams must comply with.

Activity	Date or Time Period	Recommended Actions and Deliverables
Rules Provided and Research	Fall 2025	Recommended Action(s): Review the Rules prior to Fall Kickoff.

Fall Kickoff	Fall Kickoff	Mobility Innovation Rules Committee will answer questions and provide additional guidance to teams. Recommended Action(s): Attend Fall Kickoff with questions prepared.
Idea Generation and Design	Fall Kickoff - Winter Workshop	Deliverable(s): Submit a Rules Inquiry titled "Mobility Innovation Proposal: <Team Name>". The Mobility Innovation Rules Committee will respond with a topic approval or additional questions within 10 business days. If you do not receive a response, please have your mentor reach out to MIRC. Upload a 1 page presentation summarizing each topic proposal via the Autodrive portal. If you are submitting more than 1 topic for consideration, each topic should have a separate page (i.e., a PDF with 3 pages should have 3 separate topics at 1 page each). Note: topic summaries exceeding 1 page and/or with modified formatting will be automatically denied. They can be resubmitted in the prescribed 1 page summary. Please see section "Inquiry Template" for a suggested inquiry format.
Topic Inquiries	Winter Workshop - 2 Weeks	Deliverable(s): All teams shall have their topic inquiries & topic summaries submitted.
Winter Workshop Presentation Upload	Winter Workshop - 2 Days	Deliverable(s): All teams shall upload their intended presentation 2 days prior to the start of Winter Workshop.
Winter Workshop	February 2026	Deliverable(s): Each team shall present a non-graded, 15-minute overview of their status in preparation for Summer Competition to the AutoDrive Rules Committee. This presentation replaces both the Mentor progress updates and Mentor evaluations previously held in Year III and IV. The intent of this review is for teams to demonstrate that they are prepared for Summer Competition. Teams' design solutions should be close to completion at this point.

Approved Topics Release	Winter Workshop + 1 Week	Mobility Innovation Rules Committee will release a generalized list of approved topics to Teams. Any teams still pending an approved topic can review this list to be informed of what is already approved.
Presentation Refinement	February 2026 – June 2026	Recommended Action(s): Teams presenting at Summer Competition should focus on finalizing, practicing, and honing their presentations. Teams are recommended to review and solicit feedback on their presentation with various audiences.
Presentation Upload	2 days prior to Summer Competition Start	Deliverable(s): Presentations will be completed and uploaded to the AutoDrive portal. No edits will be permitted after the submission period.
Summer Competition	June 2026	Deliverable(s): Presentations will be given live. The expectation is that a limited number of team members will present. Presentation order and times will be communicated at least 1 day prior.

MIE 1.5 Inquiry Template

Below is an image of the Inquiry Template. An editable version will be made available in Series Resources.

Team Name:

Date:

Replace With Title of Proposed Topic

Please use this space to summarize your proposed topic. Keep the summary concise. No more than one page should be used per proposal. A team may propose more than one topic/idea.

NOTE: Formatting of this document may not be changed (spacing, font size, etc.).

Example:

- 1-2 sentences contextualizing your proposed topic.
- 1-3 sentence explanation of proposed topic.

MIE 1.6 Scoring

To ensure consistency and fairness in judging at Summer Competition, a set panel of judges will score each team's presentations. Scoring of these deliverables will total 45 points.

Category	Maximum # Points / 45	Notes
Elevator Pitch	5	Captures the attention of the judges/audience, and clearly highlights the position of the team. It should include an introduction, explanation of company/product, a unique value proposition, and call to action.
Problem Statement	5	The proposal identifies and justifies a product or feature that improves the customer's AV experience.
Benefits to Stakeholders	10	The presentation should highlight the benchmarking performed by the team, provide a compelling reason for judges to invest, and highlight the benefit to the customer.
Technical Complexity	10	The selected proposal should include a technical explanation and differentiate itself from existing solutions.
Feasibility of Product/Feature	10	The proposal should be practical, financially viable, and contain a business plan.
Professionalism and Persuasion	5	The presentation is perceived as professional, the team appears as prepared to present, and judges are persuaded to invest in the proposal.

MIE 1.7 What Happens If ...?

In an effort to address questions and align rules interpretations across Teams, the following hypotheticals have been addressed in a "What happens if ...?" format:

What happens if...	Response
My team has started investigating a topic prior to MIRC approval?	MIRC reserves the right to change the team's direction. Get your topic approved early.
My team doesn't have an approved topic by the Winter Workshop - 2 weeks deadline?	Teams in this situation will receive a flat 5% penalty. Note: Teams will still need to obtain an approved topic via inquiry from MIRC if they do not have one by Winter Workshop.
My team doesn't have material ready to present at Winter Workshop?	We'll discuss presentation techniques during the designated timeslot. Teams in this situation will receive a flat 5% penalty.
My team doesn't have an approved topic before Summer Competition?	Teams will incur a 5% flat point penalty per week -- up to 20% total -- leading up to the presentation upload deadline. Penalties will be assessed by when teams submit their topic inquiry & upload their topic summary that results in an approved topic. Note: This creates a potential "grace period" for teams to submit their topic inquiry after Winter Workshop without accruing the additional 5% penalty per week.

My team wants to make changes to our Summer Competition presentation after the submission portal has closed?	No changes will be permitted. The presentation upload date has been adjusted closer to Summer Competition to address this concern. No exceptions will be permitted.
My team uploads the Summer Competition presentation late but before the first Summer Competition presentation?	A flat 20% point penalty will be assigned.
My team uploads the Summer Competition presentation after the first Summer Competition presentation?	If the presentation is not uploaded by the beginning of the first presentation at Summer Competition, the team may still choose to present but will receive an automatic 50% flat-point reduction.
My team is penalized, and I want to appeal?	Refer to MIS 1.4 Violation of the Rules. MIRC will elaborate on penalties during the final point submission.
Penalties results in a negative score?	Teams in this situation will receive a score of zero.

MIE 1.8 Penalties

The following chart summarizes penalties for this challenge. Teams that meet the prescribed deadlines and follow the rules will not incur penalties.

Flat Penalty Percentage	Flat Point Amount	Deadline	Criteria
5	2.25	Winter Workshop – 2 weeks	Topic Inquiry Due
5	2.25	Winter Workshop – 2 days	Pre-review presentation not uploaded before Winter Workshop
5	2.25	Summer Competition Upload – 4 weeks	Late Topic Inquiry Due
5	2.25	Summer Competition Upload – 3 weeks	Late Topic Inquiry Due
5	2.25	Summer Competition Upload – 2 weeks	Late Topic Inquiry Due
5	2.25	Summer Competition Upload – 1 week	Late Topic Inquiry Due
20	9	Summer Competition Upload	Presentation uploaded after deadline but before Summer Competition presentations begin
50	22.5	Summer Competition First Presentation	Presentation uploaded after first team presents

Note: MIRC will accelerate topic inquiry responses after Winter Workshop to a 3-business day turnaround. Penalties will be accrued on a week-by-week basis.

MIE 2 0-0-0: AV Minimum Viable Product (MVP) Proposal

MIE 2.1 Problem Statement

A Minimum Viable Product (MVP) is a product that contains the bare minimum content and features that produce the desired outcome for its customers. The goal of an MVP is to make considerations for resources such as time, complexity, cost, ease of implementation, etc. This approach helps in reducing risks and ensuring that the final product aligns with market demands. Any MVP must still comply with applicable regulations to be used on public roads.

MIE 2.2 Challenge & Considerations

Each team has developed unique sensor suites for your vehicle as part of the AutoDrive challenge. Your goal for this 0-0-0 challenge will be to analyze your sensor suite as intended for Year V, optimize it, and propose an MVP.

MVP Proposals shall consider:

- Benchmark Analysis
 - Compare the MVP sensor suite proposal to the team's current vehicle sensor suite and another Robotaxi AV of your choosing. Examples of Robotaxis include but are not limited to Cruise, Waymo, Zoox, etc.
- ADAS and AV Regulatory Compliance
 - Compliance to the following North America regulations: FMVSS111, FMVSS127, FMVSS208.
 - Identification and compliance to five state/locale regulations, including: Texas, California, Michigan, your team's local state/region, and one additional state/locale of your choice.
 - If your team's state is already included above (Texas, California, Michigan), you must choose an additional state/locale of your choice. The final number of state/local regulations must total five.
- Sensor Coverage
- Plan for Self-Certification to Operational Design Domain (ODD)
 - See Self-Certifications to ODD Section for further details.
- Results from MVP Optimization
 - Summary of Year V intended sensor suite to MVP proposed sensor suite.

MIE 2.3 Definitions

FMVSS: Federal Motor Vehicle Safety Standards. These are regulations set forth by the National Highway Traffic Safety Administration (NHTSA) that specify minimum safety standards for motor vehicles and related equipment.

FMVSS 111: Regulation for Rear Visibility Requirements, including equipment of rearview camera that provides a clear field of view behind the vehicle when in reverse.

FMVSS 127: Regulation for Automatic Emergency Braking and Pedestrian Automatic Emergency Braking in both daylight and darker conditions at night.

FMVSS 208: Regulation for Occupant Crash Protection design and performance of safety systems in motor vehicles. Includes seat belts, airbags, crash testing, and child restraint systems.

Self-Certification: The practice of making an official declaration that something complies with regulatory standards or procedures without independent (i.e., third party) substantiating evidence.

Operational Design Domain: A comprehensive, high-level description of the operating environment in which a system is expected to function and the functions themselves.

Note: Teams are highly encouraged to understand the complexities of each regulation. The definitions provided here are intended to be high-level.

MIE 2.4 Deliverables

Teams will be allocated 15-minutes for a live presentation at Winter Workshop. After the presentation, 10 minutes will be allocated for Q&A.

Presentations shall:

- Define the Team's intended Year V sensor suite and the identify the different functional use cases for each sensor modality.
- Propose an MVP sensor suite and compare it to the Team's intended Year V suite and one production Robotaxi of the Team's choice.
 - Teams are not limited to competition-prescribed sensors for this analysis, nor are sensors required to be automotive grade. The MVP sensor suite does, however, need to meet functional requirements as Teams derive them from the ODD and regulations.
 - Comparisons should include BOM cost and an estimated power consumption in kWh.
- Explain how the MVP sensor suite will comply to the prescribed ODD and federal, state, and local regulatory requirements. This should include:
 - MVP Sensor coverage map comparison.
 - Analysis of ODD requirements to sensor capabilities.
 - Identification of regulatory requirements and analysis of sensor capabilities for compliance.
 - Proposed self-certification plan.
- Any challenges and/or key learnings.

Finally, teams are to upload their presentation 2 days prior to the Winter Workshop. If teams elect to prepare a separate document with supporting materials, that may also be uploaded to the AutoDrive document portal.

MIE 2.5 Timing and Enablers

The following is a high-level project management plan for this challenge. Note that there are deliverables that all teams must comply with.

Activity	Date or Time Period	Details, Deliverables, and Recommendations
Rules Released	Fall 2025	Recommended Action(s): Review the Rules prior to Fall Kickoff.
Fall Kickoff	Fall Kickoff	Rules Committee will answer questions and provide additional guidance to teams. Recommended Action(s): Attend Fall Kickoff with questions prepared.
Winter Workshop Presentation Upload	Winter Workshop - 2 Days	Deliverable(s): All teams shall upload their intended presentation 2 days prior to the start of Winter Workshop.
Winter Workshop	February 2026	Teams will deliver their presentations.

MIE 2.5 Scoring

Scoring will be based on the presentation and Q&A from a set panel of judges. Scoring of these deliverables will total 30 points based on the following:

Category	# Points / 30	Notes
Year V Sensor Suite Explanation	3	Effective summary of Year V sensor suite with functional use cases of each sensor identified.
Benchmark Analysis	5	Quality comparison of MVP sensor suite proposal to current vehicle sensor suite and another Robotaxi AV.
Proposed MVP Solution	15	Proposed solution includes sensor coverage map, analysis of how MVP solution meets ODD and regulations, and proposed plan for self-certification.
Technical Depth & Accuracy	4	The deliverables are technically sound, and a thorough analysis is evident. The MVP sensor suite is technically accurate and feasible.
Professionalism	3	The presentation is well prepared and professional.

MIE 2.6 Self-Certification to ODD

2.6.1 ODD Definition

The Operational Domain Definition for this challenge is as follows:

Category	ODD Type	Details
Weather	Environment	Day, Night, Light Rain, Light Snow, Dusk, Dawn.
Geolocation	Environment	All roads in USA and Canada.
Road Types	Environment	Paved and gravel roads, intersections, roundabouts.
Road Grade	Environment	0 +/- 37% grade.
Congestion	Environment	Light traffic to traffic jams.
Parking	Function	Park in residential driveways. This includes parallel, perpendicular, and angled parking in both forward and reverse gears.
Object Avoidance	Function	Avoid objects > 625cm ² (25cm x 25cm) in detectable area in all driving scenarios up to distance of 10cm from vehicle dripline.
Longitudinal Controls	Function	Maintain headway distance in 0-85mph speed range.
Longitudinal Controls	Function	Identify and adjust vehicle speed based on traffic sign and speed limit sign inputs.
Path Planning and Execution	Function	Determine and adjust route based on internal navigation and traffic conditions.
Path Planning and Execution	Function	Identify and appropriately react to traffic signs and traffic lights.
Path Planning and Execution	Function	Identify and appropriately react to active emergency vehicles.
Lateral Controls	Function	Maintain lane centering, automatically change lanes when necessary, and turn at intersections.

2.6.2 Self-Certification

Self-certification plans include several key steps:

- Responsibilities: Identify responsible members for ensuring the adherence to standards, conducting tests and evaluations, and reviewing and validating the test results.
- Compliance Assessment: Identify the specific standards that must be compliant and develop a checklist of requirements.
- Test definition and analysis methods: Define test methods and data analysis methods that address compliance needs.
- Documentation: Define methods for compliance assessment record management and updates for any delta testing.

MIE 2.7 Penalties

The following chart summarizes penalties for this challenge. Teams that meet the prescribed deadlines and follow the rules will not incur penalties.

Flat Penalty Percentage	Flat Point Amount	Deadline	Criteria
20	6	Winter Workshop – 2 days	Presentation uploaded after Winter Workshop – 2 day deadline.
50	15	Winter Workshop First Presentation	Presentation uploaded or edited after first team presents.

Article D6: MathWorks Simulation Challenge Year 5

MSC.1.1 Challenge Requirements

- Submission Format:
 - PowerPoint Submission before the GM/SAE AutoDrive II Year 5 Winter Workshop
 - Video Submission before the GM/SAE AutoDrive II February 1st
 - PowerPoint Presentation at GM/SAE AutoDrive II Year 5 Winter Workshop
- Presentation Duration: 30 minutes
- Presentation format:
 - The presentation format allows for a maximum of 3 presenters, all of whom must be undergraduate students
 - Failure to comply with this requirement will result in a 20% deduction from the total available score.
- Video Requirement:
 - This year we are adding an additional requirement of producing a summary PR video of what you have done in the past 5 years. More details of the requirements of the video are included in the section below.
- Presentation requirement:

It is important to note that the presentation requirement is established to facilitate judges in following along with your presentation and providing valuable feedback. Failure to adhere to these requirements may pose challenges for judges when evaluating your work.

 - All presentation materials, including videos and slide decks, must be submitted to the portal prior to the start of the AutoDrive Challenge II Year 5 Winter Workshop.
 - Presentation slides must include slide numbers for easy reference.
 - It is also necessary to clearly label each slide and indicate the corresponding section it belongs to.
 - The first slide should introduce the simulation team composition and provide a brief (less than 1 minute) introduction to the presenting members.

MSC.1.2 Challenge Objective

- The goal of the simulation challenge is to prepare students to understand the importance of simulation in automotive software engineering, showcasing best practices similar to the industry, which will be highly valued.
 - MathWorks core value: Innovative Model-Based-Design simulation
 - GM's core value: Zero Crashes, Zero Emissions and Zero Congestion
- In your presentation, be sure to include videos, pictures, gifs, and screen captures wherever appropriate (such as results, algorithm performance comparison, scenario creation, metrics evaluation, etc.). Live demos are not recommended.
- We expect you to perform both qualitative and quantitative analyses for these sections to support your observations and results. Qualitative analysis is very important and being judged heavily.
- The use of the MathWorks Tools is required to perform the challenge. You are welcome to introduce additional tools to supplement the MathWorks tooling.
- In Year 5 of the simulation challenge, you will use the automated simulation system you have developed over the past 5 years to verify to enable more frequent testing during the early stages of software algorithm development and validate your software stack. The simulation challenge supports the development of dynamic events for the grand finale of the AutoDrive Challenge II, helping you gain practical experience and insights of your system performance.

- As a simulation challenge, we are more interested in seeing you describe to us how you test your algorithm. So, focus on how you used simulation tools for algorithm testing and validation.
- Keep in mind if your simulation has identified problems in your software, it's a good thing! It means you are doing what software simulation is intended to do, and you should document it and share it in your presentation.

MSC.1.3 Challenge Scoring

SECTION		POINTS
1	Automated Simulation System Showcase	20
2	Testing Strategy and Scenario Variation	20
3	Controller Robustness Testing: Snowy Weather	20
4	Reflection on Simulation Challenge in Series II	5
5	Video Summary of ADC II Simulation Challenge	10
Total		75

This year's finale will differ slightly from previous editions. We're shifting the focus toward a more holistic reflection of your simulation system and testing strategy.

- We expect each team to put effort into finalizing and integrating your simulation solution into your daily development workflow. If your simulation uncovers abnormal behavior, what steps do you take? How do you act on that data?
- We also want you to showcase your overall testing strategy and the scenarios you've developed. What infrastructure have you built to ensure software quality? What kinds of scenario setups do you use to test different features? How do you link specific features to specific scenarios?
- Finally, we challenge you to go beyond what MCity provides. How robust is your vehicle controller under edge-case conditions? Can your vehicle handle unexpected inputs or challenging environments like a slippery road? What limitations or edge cases have you identified in your current system?

You are strongly encouraged to go beyond the provided challenge description by exploring additional features and functionalities in your simulation. This is your opportunity to showcase your software development and testing skills in a creative and impactful way.

Keep in mind: 25% of the points in each section are reserved for "Innovation" and "Judge's Discretion." Use this as a chance to demonstrate your innovative use of the MathWorks Simulation suite and go above and beyond the baseline requirements.

To earn full points in a section, your submission must:

1. Thoroughly address all bullets listed in the scorecard for that section.
2. Effectively complete any highlighted tasks or show clear innovation in your approach.

3. Impress in the “Judge’s Discretion” category through high-quality content and/or meaningful work that extends beyond the listed requirements.

MSC.1.3.1 Automated Simulation System Showcase [20 points]

This section evaluates how well your simulation system is integrated into your team’s development workflow and how actively it is being used. We are looking for more than just simulation results — we want to see a functioning system that supports your engineering process from code to insight. Treat this as an opportunity to showcase the maturity and practicality of your infrastructure.

A. Simulation System Architecture (8 points)

This is the most heavily weighted part of the section. You should clearly demonstrate how your simulation system fits into your overall development cycle. What does the end-to-end process look like, starting from a developer pushing new code to receiving simulation test results? Describe the tools, services, and workflows you’ve established to support this. We’re especially interested in seeing structured, repeatable processes rather than ad hoc usage. A great way to demonstrate this is to include a short video or GIF showing the full pipeline in action: For example, a developer pushing to Git, triggering an automated simulation run, and receiving results on a dashboard or in logs. This shows that your system is being used and trusted as part of your daily workflow.

B. Automation For Executing Simulation (4 points)

Here, we want to understand how your simulations are executed in practice. Are they triggered manually, scheduled to run nightly, or automatically kicked off by a CI/CD pipeline? Describe your level of automation, and whether you’ve built systems for job scheduling, batch processing, or parallel runs. Be specific about how often your simulations are run and how much they scale. Evidence of consistent, automated runs such as screenshots of CI jobs, pipeline logs, or summaries of past simulation results will help demonstrate that your system is robust and actively maintained.

C. Simulation Result Feedback & Handling Abnormal Results (3 points)

Your simulation system should not operate in isolation. It should close the loop with your development team. In this part, explain how simulation results are communicated back to developers. Do results get posted to a dashboard, emailed to the team, or flagged in issue trackers? When something abnormal happens in a simulation such as a system failure, unexpected behavior, or a failed test, how does your team detect, investigate, and respond to it? You should share examples of how this process has played out during development: a specific bug that was caught via simulation, how it was triaged, and how it informed improvements in your code or test coverage. This shows us that your system is designed to be actionable.

D. Innovation & Judge’s Discretion (5 points)

Up to 25% of the score in this section is reserved for innovation and creativity. This is where you can stand out by going beyond the expected. Maybe you’ve created a dashboard, built a smart way to prioritize scenario tests, or integrated fault injection or uncertainty modeling into your workflow. We’re looking for thoughtful, technically sound innovations that enhance your system’s capability or usability. Whether it’s through advanced infrastructure, smart developer tooling, or creative extensions of the simulation platform, this is your chance to surprise and impress.

MSC.1.3.2 Testing Strategy and Scenario Variation [20 points]

This section focuses on how well your team designs and uses simulation scenarios to evaluate software functionality. We’re looking for structured, feature-driven testing that uses varied conditions, measurable outcomes, and continuous refinement based on real-world findings.

A. Scenario Variation & Feature Mapping (5 points)

We want to see how well your scenarios are designed to test specific software features under varied conditions. For each major feature such as perception, planning, or re-routing, describe how your simulation environment is used to validate it. Strong teams will show how scenario parameters (e.g., road layout, obstacle type, timing) are varied automatically or systematically to test edge cases and robustness. For example, dynamic re-routing might be tested with roadblocks placed in different positions across multiple runs. We are especially interested in whether each scenario has a clear purpose and connection to a specific system behavior. Show us a list of features your simulation supports to test.

B. Use of Metrics to Evaluate Software Behavior (5 points)

Good simulation testing requires quantifiable outcomes. Describe what metrics you use to evaluate whether a scenario passes, fails, or reveals performance gaps. These might include task completion rate, time to recovery, trajectory smoothness, distance to obstacles, or system latency. Explain how you chose these metrics and how they align with your feature requirements. We're also looking for evidence that your team tracks these metrics over time, uses them to guide development, or even gates code changes based on simulation results.

C. Scenario Creation & Regression Tests Creation (5 points)

A mature simulation workflow evolves over time. If you encounter bugs or edge cases in physical testing, competition, or even previous simulations, can you recreate those as new simulation scenarios? Describe your process for identifying high-value or failure-inducing situations and integrating them into your test library. We want to see that your simulation environment supports regression testing and continuous improvement, not just one-off evaluations.

D. Innovation & Judge's Discretion (5 points)

This category rewards creative, technically sound enhancements to your scenario-based testing. This could include automatic scenario generation, test prioritization based on recent code changes, uncertainty testing, or advanced visualization of test outcomes. Any element that goes beyond the basics and demonstrates depth in your testing strategy may be recognized here.

MSC.1.3.3 Controller Robustness Testing: Snowy Weather [20 points]

This section challenges your team to evaluate how well your vehicle controller handles low-traction environments, such as snowy or icy roads. Using the GM and MathWorks-provided Bolt EUV model, you are expected to build simulations that expose your controller to challenging surface conditions and demonstrate that your system maintains safe and stable behavior.

A. Integration of the Bolt EUV Model & Simulation Setup (5 points)

You must demonstrate that the Bolt EUV model has been successfully integrated into your simulation environment. This includes configuring the model appropriately and simulating realistic slippery conditions through surface parameters, weather models, or environmental adjustments. Describe how traction degradation is modeled and how the simulation setup creates a believable snowy driving environment.

B. Scenario Design: Low-Traction Events (5 points)

Create multiple scenarios that challenge the controller with different types of low-traction conditions. These might include lane changes on icy roads, braking on downhill slopes, or evasive maneuvers on patchy snow. The goal is to go beyond a simple "slippery road" and demonstrate thoughtful coverage of how adverse weather impacts driving behavior. Be specific about how the scenarios were designed and how they relate to real-world edge cases.

C. Controller Robustness & Analysis of Limitations (5 points)

This part evaluates how your vehicle controller responds to low-traction challenges and whether it maintains safety and stability. Does it detect slipping or instability? Can it adjust trajectory or braking to recover? What metrics do you use to assess performance — such as lane deviation, stopping distance, or recovery time? Importantly, show us not only where the controller succeeds, but also where it struggles. Identifying failure cases or instability modes shows you've critically evaluated system behavior and are thinking ahead to future improvements.

D. Innovation & Judge's Discretion (5 points)

Up to 25% of the score in this section is reserved for creative approaches, advanced control strategies, or thoughtful test design. This could include adaptive control systems that adjust for surface conditions, innovative scenario generation, integration of real-time surface detection, or clever metrics and visualizations. Teams that go beyond the prompt with technically sound and well-documented ideas will be rewarded here.

MSC.1.3.4 Reflection on Simulation Challenge in AutoDrive Challenge II

As we reach the final stages of this five-year journey, we ask you to take a step back and reflect on your experience in the AutoDrive Challenge II Simulation Challenge.

What are the biggest lessons your team has learned across the years? If you were to start over, what would you do differently in terms of process, collaboration, or technical design? Reflect not only on your successes, but also on the challenges, missteps, or pivots that shaped your approach.

From a tooling and infrastructure perspective, we'd also like your feedback on the competition itself. What could have made your work more effective? Are there simulation tools, scenario libraries, datasets, documentation, or infrastructure that would have helped you build better systems faster? What should future versions of this challenge do differently?

MSC.1.3.5 PR Video: Video Summary of Your Team's ADCII Simulation Challenge

As part of your final submission, your team must produce a 3-minute video that showcases your simulation infrastructure and demonstrates how simulation is used to ensure software quality in your development process. This is not a technical tutorial — it is a PR-oriented highlight video meant to promote your team's work, your process, and the broader impact of simulation in autonomy development.

Think of it as a simulation showcase video you would share with a sponsor, advisor, senior executives from GM and Mathworks, or public audience. It should clearly convey what your simulation system does, how it supports software development, and why it matters — all in a format that's professional, polished, and engaging.

Your video should cover:

- The end-to-end simulation workflow: from developer code to test results
- How simulation supports daily development, testing, and validation
- How your team ensures safety, quality, and robustness using simulation
- What sets your system apart or makes it scalable, reliable, or innovative

This is not just a voiceover on a screen recording:

We expect a thoughtfully edited video that blends visuals (e.g., scenario replays, diagrams, result dashboards, team clips, or live-action footage) with a clear story or message. Use voiceover, music, captions, or interviews — whatever helps tell your simulation story to a broader audience.

Scoring

This section will be ranked competitively across all teams:

Place	Point Value
1 st Place	10 points
2 nd Place	9 points
3 rd Place	8 points
4 th Place	7 points
5 th Place	6 points
6 th Place	5 points
7 th Place	4 points
8 th Place	3 points
9 th Place	2 points
10 th Place	1 point

Judges will evaluate videos based on clarity, communication impact, production quality, and how well they represent your team's simulation-driven approach to autonomy.

MSC.1.4 Challenge Scorecard

AutoDrive II Year 5 MathWorks Simulation Challenge		
Students: Please refer to the simulation challenge EOD document for detailed instructions!		
Section	Description	Points
AUTOMATED SIMULATION SYSTEM SHOWCASE 20 pts	A) Simulation System Architecture	8
	B) Automation For Executing Simulation	4
	C) Simulation Result Feedback & Handling Abnormal Results	3
	Judge's Discretion: Has the team gone above and beyond in this section?	5
TESTING STRATEGY AND SCENARIO VARIATION 20 pts	A) Scenario Variation & Feature Mapping	5
	B) Use of Metrics to Evaluate Software Behavior	5
	C) Scenario Creation & Regression Tests Creation	5
	Judge's Discretion: Has the team gone above and beyond in this section?	5
CONTROLLER ROBUSTNESS TESTING: SNOWY WEATHER 20 pts	A) Integration of the Bolt EUV Model & Simulation Setup	5
	B) Scenario Design: Low-Traction Events	5
	C) Controller Robustness & Analysis of Limitations	5
	Judge's Discretion: Has the team gone above and beyond in this section?	5
REFLECTION ON SIMULATION CHALLENGE IN SERIES II 5 pts	REFLECTION ON SIMULATION CHALLENGE IN SERIES II	5
PR Video 10 pts	Scoring based on ranking achieved	10
Total Points		75

Article D7: Project Management Year 5

PM.1.1 Project Management Objective

The topic of Program Management (PM) goes by many names in various industries: Project Engineering (PE), Project Management (PM), Product Management (PM), Program Management (PGM), etc. To familiarize you with these terms, we will use them interchangeably.

The goal of the Project Management challenge is to have the teams use tools to better manage and execute their plan for the AutoDrive challenge.

For Year 5, the structure of this challenge has evolved and changed. **The five core principles from previous years have been combined to become 4 core principles (Team Structure and Team Communication have been combined). Also, the Winter Workshop format has shifted from a written report to a live demonstration / presentation of these principles with Q&A.**

The Winter Workshop will now require teams to present their four project management tools—Gantt Chart, Work Breakdown Structure, Team Structure and Communications, and the Decision-Making Model, along with a Summary — in a live presentation format. It is expected that the PM will pull up a live view of the tool being discussed (i.e. during the Gantt discussion, the live Gantt chart and Critical Path should be pulled up and displayed, same for WBS). Judges will evaluate both the content and the team's ability to clearly communicate their approach and exemplified proficiency of each tool. Each section will have a 5 min presentation with a 5 min Q&A about that section along with a summary 5 min presentation and an extended Q&A after that. This will be scheduled to be 1 hour long.

The level of detail and rigor expected in each tool remains unchanged from previous years. Teams should not assume that the presentation format makes the deliverables easier; in fact, expectations for clarity, completeness, proficiency, and professionalism are higher.

Analyze and summarize the following for EACH deliverable in your winter workshop **presentation**:

- Show the deliverable live – i.e. pull up your real Gantt chart, Decision Making Model, etc.
 - You could use snips in a PowerPoint for the presentation to highlight a certain event but the live version must be up and active so it can be referenced for the judges during Q&A
- Compare what you did last year to what you have for this year. What continued, what was deprecated, what is new? Why did you keep it or why is it different? Basically, what did you change and why?
- Evaluate the tools you used. Are they the same as last year? Why / why not?
- What went well? What didn't go well?
- Based on last year's performance, what did you change in PM to address any & all shortcomings?

Format: 1 hour Time Slot: You will walk through each deliverable pausing between each for judges questions. You can use up to 5 minutes to present your deliverable before the judges stop you for questions. You do not need to use the full 5 minutes for each. However, some of these sections could be challenging to discuss in detail in 5 minutes. So, practice and focus on the rubric deliverables.

- Intro- up to 5 minutes
- Gantt Chart -up to 5 minutes
 - Judges questions – 5 minutes
- WBS- up to 5 minutes
 - Judges questions 5 minutes

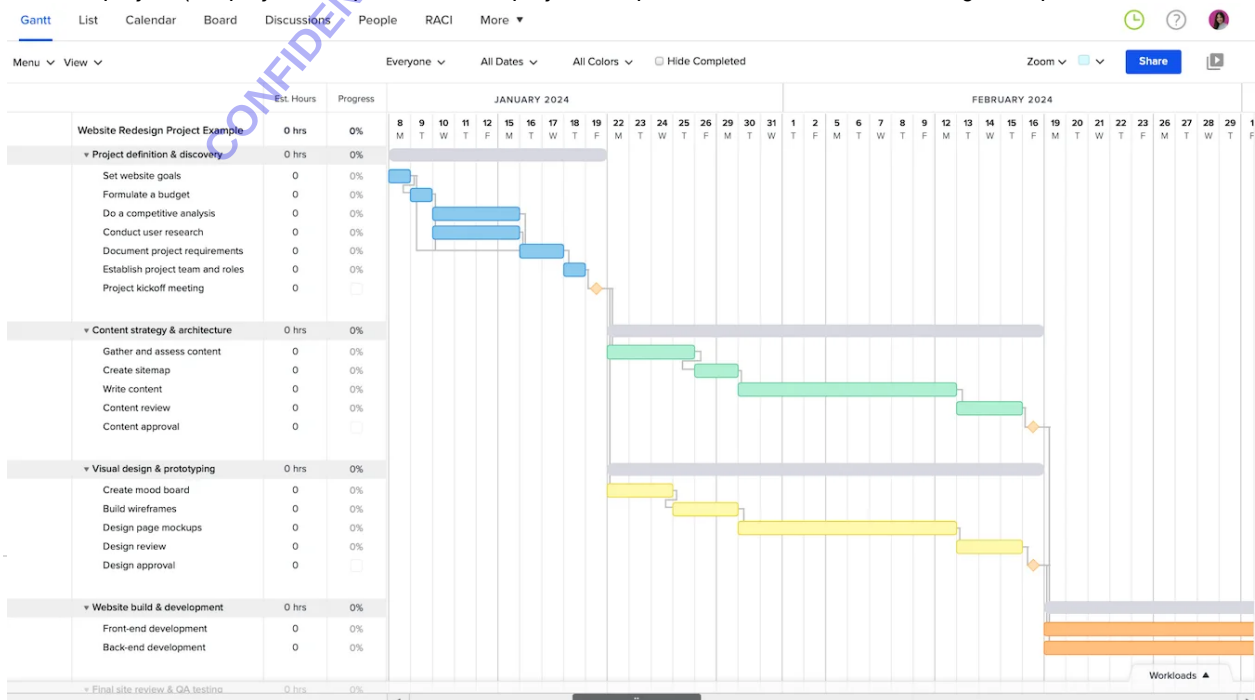
-Continued for other deliverables.....
- Summary – up to 5 minutes
 - Final comments/questions from the judges

At the Final Competition, teams will submit a written report (same format as Year 4) with an added feedback section, and deliver a presentation that reflects back on their project management journey / team journey over the past 5-years.

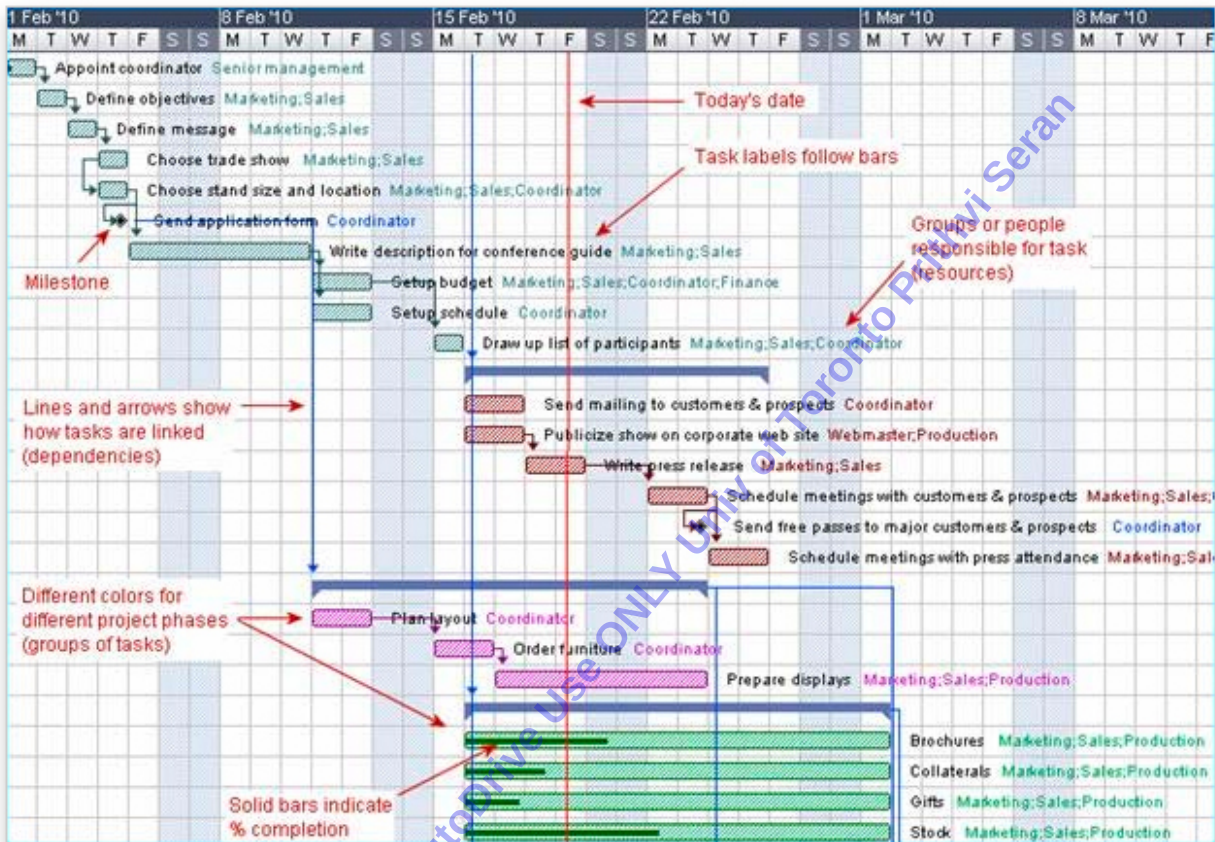
PM.1.2 Project Management Deliverables

PM.1.2.1 Gantt Chart

The first goal of a successful Project Engineer is to understand the scope and timing of the project. The easiest way to view this is with a Gantt chart, which is sometimes called a bar graph timing chart. This is a very simple concept where the deliverables are considered single date milestones (typically represented with a diamond symbol) and the bar graphs represent the time expected to complete the work associated with that milestone. Gantt charts and Work Breakdown Structures (WBS) work hand in hand and are sometimes interchangeable in meaning and descriptions. WBS will be covered in the next section. Some of the milestones will be obvious from the other parts of the AutoDrive Challenge's deliverables (all AutoDrive Deliverables, workshops, and challenges should be Gantt Chart milestones), and some milestones will be up to the individual teams to mark interim goals or review points. Usually, each milestone will require multiple different work streams (bar graphs) to be completed to successfully achieve the milestone. These can be sequential efforts, and these can be parallel efforts. Most often having a combination of parallel and sequential activities required to complete a single milestone. Understanding these relationships in a visual representation is the goal of the Gantt chart. Typically, the milestones will be represented as a diamond shape and the work element bar graphs will be connected as they cascade via arrows to indicate required predecessors and successors or what is commonly called relationships. Connectivity of all tasks is imperative and must be 100% complete; every task must be linked and cascaded down to a milestone. [Even tasks that appear to be a dead-end task (i.e. turn in report), must cascade down to the next or final task of the project. The logic here is that the project would not be complete if that didn't happen.] Then milestones can be cascaded and linked to other tasks or milestones. Basically, every single item in the Gantt chart should be linked to following items, with the only exception being the final milestone of the project (i.e. project competition and/or project complete. Please see the following examples.



Gantt Chart Example 1

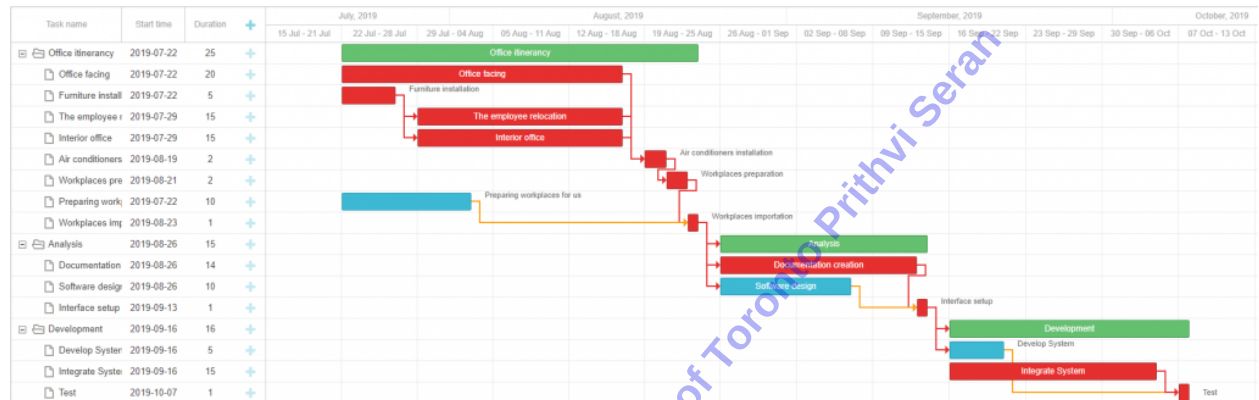


Gantt Chart example 2

These 2 examples are very simple projects just intended to give a visual reference. Your actual project timeline should be significantly larger and cover 5 epochs, or what will be the 5-year challenge. It is expected that each year will have its own visualization of a Gantt chart (a stand-alone Gantt chart), but these can be linked together to show a 5 year-long project that achieves the final goals. (Of course, this may not be a continuous Gantt chart if the Gantt chart tool has changed significantly over the 5 years; changing tools will create a natural break in the 5-year project Gantt.) It will be expected that at least year 5 is very detailed, laying out each individual workstream. All long-term projects work this way. As you move forward in time, more information is learned, and the Gantt chart gets updated to reflect the current level of information. Therefore, Gantt charts should be living documents that grow and get modified constantly. The key to good Project Engineering is accurately tracking these changes and understanding why these changes are occurring.

One of the most important parts of a well-constructed Gantt chart is highlighting and understanding the "Critical Path". The critical path is the most "at risk" chain of events that are driving key (critical) project

milestones. This is typically the chain of tasks that is the tightest packed that are critical to the key deliverables. Highlighting and tracking the critical path is very important to good project engineering. Critical paths are not the entire project! The Critical path should be a small section of the entire Gantt. Critical paths are the tasks that, if they slip at all, then the project is delayed or late. The Critical Path should also be shown on the Gantt chart and is typically shown as red, to indicate that this is the Critical Path. See Fig 3 below.



Gantt Chart Example 3 (Critical Path)

Also note that the Critical Path can span across sections of the Gantt Chart, it is possible that one section will be a requirement or predecessor of another section, and those two interactions could be on the Critical Path. A Critical Path analysis is one of the key elements or analysis of a good Project Engineer.

A good Critical Path analysis will include: What are the key tasks that are driving the project timeline? Are these key tasks on time, or are they late? If they are late, why? Also, if they are late, what is the team doing to get back on track? What is the team doing to ensure that no further slips occur? What could have been done differently to prevent this in the future? How has this impacted the Work Breakdown allocation of time? Has the Critical Path stayed the same, or has other elements now joined or become part of the Critical Path, remember a Gantt Chart is a living artifact that is constantly being updated! It is important that not only is the Critical Path identified but is discussed and analyzed.

Year-5 will be different than the previous years. This year, the Gantt chart will be evaluated in depth only once during the Winter workshop. The Year End Challenge will just review any changes that occurred. Even though there will be no written document for the Winter Workshop, it is expected that the teams will still maintain the Gantt chart and Critical Path and are constantly monitoring these and making adjustments to reflect reality. The Year End deliverable will be discussed in a later section.

Note: There are many Gantt chart tools (WBS tools) available online that work significantly better than just doing this in Excel. It is highly recommended that each team acquires a Gantt chart tool that also ties the WBS elements and Gantt chart together.

PM.1.2.2 Work Breakdown Structure

The WBS or Work Breakdown Structure is part of the Gantt chart from the previous section. A project timeline Gantt chart without appropriate WBS assignments is meaningless. A simplistic view of these two

would state that: 1. The Gantt chart is a list of when things need to be done (milestones) and what work is required to meet these milestones (this is where the WBS and Gantt charts overlap). 2. The WBS is a list of what will be or is being done and, more importantly, what resources are required to do that work. For most WBS elements this will be the people doing the work and tracking their time. The WBS is a tool meant to help plan and optimize resource utilization. If used effectively, it should be incorporated in planning what resources will be used for a task prior to beginning work. The WBS should also be regularly monitored during task execution to ensure that resources are appropriately utilized. For example, if person #1 can only work on a project for 8 hours a day but only 2 days a week and person #2 can work on a project 2.5 hours per day but just 3 days a week, person #1 can produce 2 times the work as person #2 (assuming the skill set and work level is the same). This can also be helpful in balancing workload for members of the team. The WBS can help identify if a specific person is being over or underutilized. This is not meant to be a timecard but as a resource that the project management team can leverage to assist in forecasting and balancing workload. The WBS combined with the Gantt can assist with planning work needed from members of the team and how many people will be required to reach a deliverable. This all gets reflected in the WBS.

There are many different ways of tracking WBS elements. AutoDrive will not force a particular method. These are dictated usually by the software being used.

The key points will be:

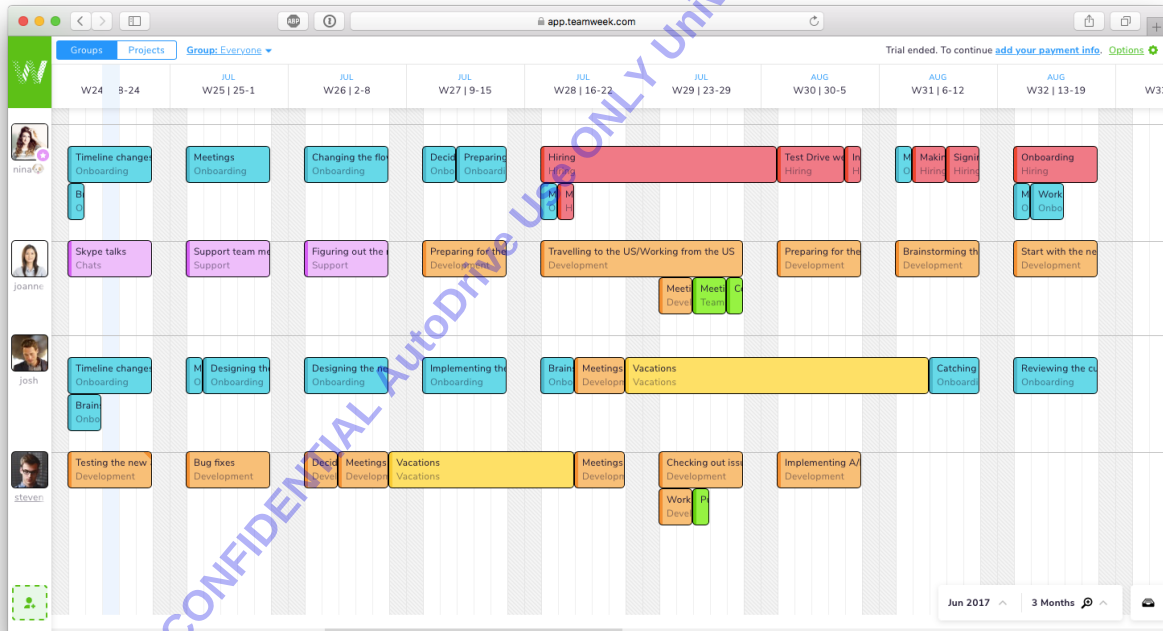
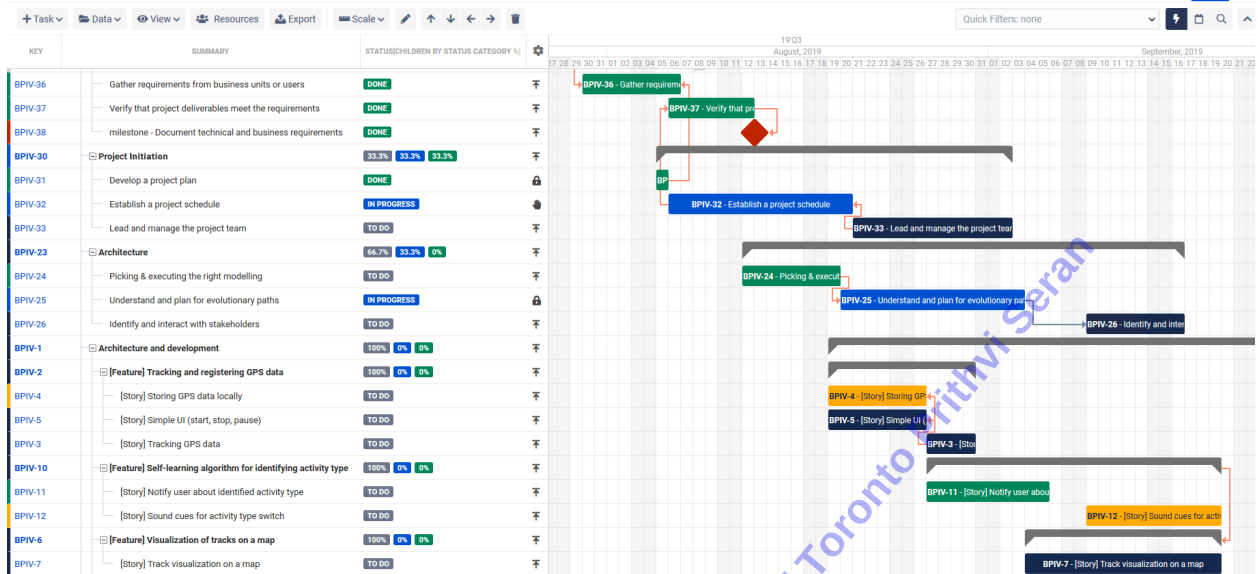
- Comprised of a comprehensive deliverable list
- Has individual names/group names assigned to each work package, if group names are used, then those groups must list all team members (required for last bullet).
- Includes hrs/wk per person
- Is tied to the Gantt Chart bar line items (line by line)
- Needs to be formatted in a way that it's easy to visualize and balance individual's workloads. Includes an overview of each team member's workload, shown week by week in hours/week
- Evidence the WBS was utilized during deliverable planning

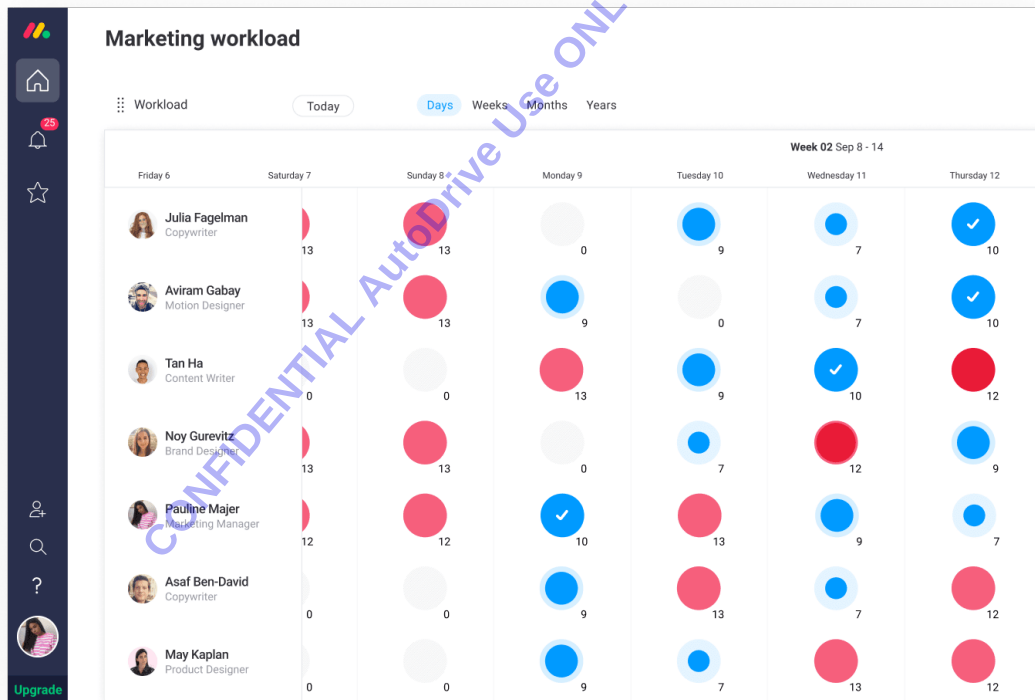
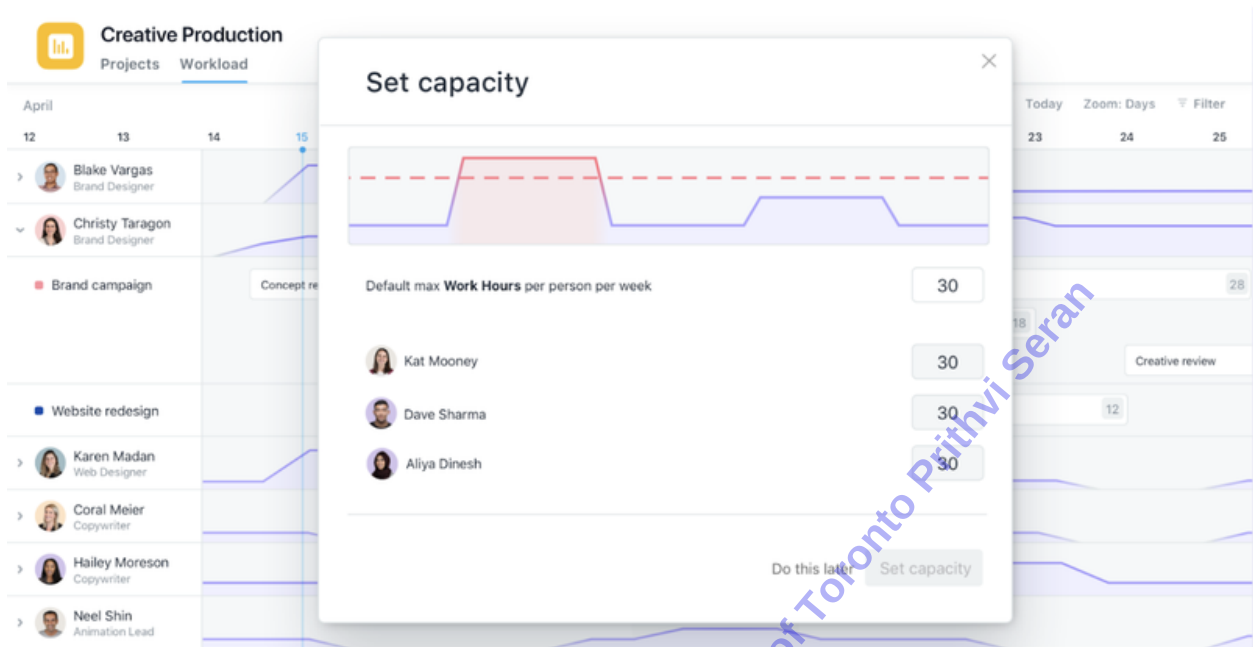
Like the Gantt chart, the WBS is a living document, constantly tracking changes. The WBS documentation will be required at each of the previously mentioned Gantt chart deliverables.

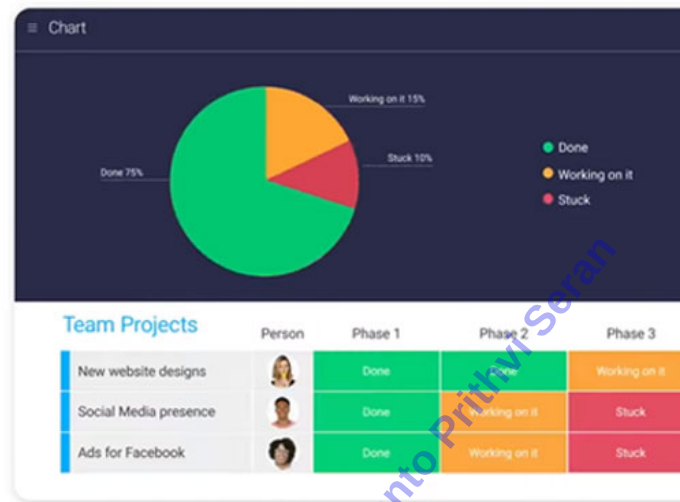
As stated before, the WBS deliverable for Year-5 will be different than previous years. The entire WBS will only be evaluated once during the Winter workshop. Even though there will be no written document for the Winter Workshop, it is expected that the teams are utilizing the WBS for planning and will consistently monitor it during deliverable execution to make adjustments. The judges will be looking for evidence of both aspects. The Year End deliverable will be discussed in a later section.

Following are examples of WBS.

The most popular Gantt chart for Jira







PM.1.2.3 Team Structure & Communication Plan

Having a team structure is important for determining who is ultimately responsible for different areas of the project and can help newer students on the team figure out how the team is run. Having a communication plan is important for keeping everyone on the team informed on what is happening within the team and for ensuring important data is not lost. For this section you will need to define your team's roles, structure, and communication plan.

All the possible roles on your team should be listed along with the responsibilities of each of those roles. Indicate how a person can attain each role such as being elected to the role, being given the role by a leader, earning the role through completing tasks, being recruited to the role for having a particular skillset, etc. Also include what (if any) requirements exist for gaining and maintaining a role and how long each role will last.

An organizational chart should be created that shows how the defined roles interact with each other and who fills those roles. It should also show how sub teams are coordinated with the team as a whole and who the team's project manager and other team leaders are. Ultimately, this chart should include every person that is on the team. In some cases, multiple people can have the same role (i.e., everyone on a sub team might have the same sub team role) or one person can have multiple roles (i.e., the Project Manager might also be the Chief Pizza Officer). Care should be taken to make sure that one person does not have too many roles. Please indicate what degree and grade level of education each member of your team is currently working towards and number of years they have been involved in AutoDrive (including if student participated in AutoDrive series 1). For example, John Smith Sophomore B.S. Electrical Engineering, AutoDrive series 2 years 2-3.

There are many different types of organizational structures that are commonly used in business such as line staff, matrix, projectized, etc. You are free to choose whichever organizational structure you think will fit best with your team and what roles you will have on your team (there is no set template). However, you are required to always have a "project manager" role whose responsibilities include at a minimum the project management deliverables. If your project manager leaves the team an interim project manager must immediately be named until a permanent replacement can be found. Supporting individuals that are not directly on the team such as faculty advisors, industry advisors, lab technicians, and others can also be included in the roles and organizational chart, however, as it is a

student run project their responsibilities should be limited to advising the team only and not include decision making, completion of tasks, etc.

Create a communication plan for your team that answers the basic questions of how you communicate within the team that can be referenced by your team members if they are unsure where/who to share information with or where to find information. Your plan should include informal, formal, documented, and external communications.

For informal communications describe how day-to-day communication is conducted between individual members of your team or smaller groups on your team. Any platforms used by the team should be listed (i.e., messaging apps, team chat software, texting, email, snail mail, social media, Kanban board, etc.). If multiple platforms are used describe how you choose what is communicated on each platform. Your formal communications can include communications that are sent to the full team, any official team meetings, and team schedules. List out what platforms are used for your team's formal communications (these could be the same as informal) and what information is planned to be communicated on those platforms. Also create a schedule that shows reoccurring meetings and any designated work times which could include open lab time, test track time, sub-team work time.

Documented communications are any information or knowledge that needs to be stored long term. This can include software / code, simulation models, CAD models, submitted documents, lessons learned, test data, etc. Describe your change control process and document retention plan for any information or data that needs to be stored. Include information on what tools are used (Git, SVN, network drives, etc.), who has access to those tools, and what (if any) review process is done to update those files. Provide examples/proof of your change control process being used throughout the year.

External communications are any communications between the team and persons outside of the team. These can include recruitment efforts, sponsor engagement, and public communications. Describe in detail how you recruited new students to be on your team. This could include efforts to recruit students with specific talent or more general recruitment efforts. For sponsor engagement describe how you communicated with your existing sponsors and what (if any) efforts were made to recruit new sponsors. List out any public social media channels that your team has and describe what each is used for. If your team is involved in any other public communications such as attending community events, television interviews, or something similar describe what was done and how it benefited the team or your community.

PM.1.2.4 Decision-Making Model (DMM)

A decision-making model is an important tool for teams to use because it shows how information should flow and where decisions get made. This model helps teams define who needs to be involved when making a decision, and what decision can be made individually/without the entire team. The key to this is finding a process that doesn't slow your progress down because every decision needs to be vetted through a large number of people, but also making sure that decisions that affect other parts of the system are being communicated.

Your decision-making model should tie into the team structure, showing each sub-team. Team members should be able to reference this chart at any point to help them understand: (1) who needs to be involved in the decision-making process for a specific item and (2) who gets the final say in the decision. Often, in the corporate world, a DMM differs from the organizational chart. However, in this case, it's possible that your organizational chart is the same as the DMM. Decisions are not only technical decisions but also team management decisions- i.e. who should fill a specific role or do a specific task?

Along with creating DMM, your team should be able to explain it and how the team uses it. This should include who has the final say in all disputes and who reconciles sub team debates. If your DMM is the same as your organizational chart, indicate the levels of decisions. Which subgroups can make which decisions and how do these subgroup decisions get aggregated into the overall project? If your DMM is different from your organizational chart, please explain and elaborate on any differences.

You must show a visual representation of the complete decision-making model. The team should include examples of how the model was used throughout the year.

PM.1.2.6 Summary

This section applies only to the Final report and presentation. The summary should provide a high-level overview of the teams' year 5 performance using PM terms. Describe in detail the health of the team and how you feel the team is doing. Explain what went well and did not go well for year 5. Additionally, the team should include how they did over the five years of series 2. Was the team satisfied with their performance, why or why not? Looking back, what would the team have done differently if you could?

PM.1.3 Project Leadership Format and Judging

PM.1.3.1 Scoring Process

Winter Workshop

Your project manager (or the interim) must be in attendance at the winter workshop and be the main presenter.

The Winter Workshop will now require teams to present their four project management tools—Gantt Chart, Work Breakdown Structure, Team Structure and Communications, and the Decision-Making Model, along with a Summary — in a live presentation format. It is expected that the PM will pull up a live view of the tool being discussed (i.e. during the Gantt discussion, the live Gantt chart should be pulled up and displayed, same for WBS). Judges will evaluate both the content and the team's ability to clearly communicate their approach and exemplified proficiency of each tool. Each section will have a 5 min presentation with a 5 min Q&A about that section along with a summary 5 min presentation and an extended Q&A after that. This will be scheduled to be 1 hour long.

The level of detail and rigor expected in each tool remains unchanged from previous years. Teams should not assume that the presentation format makes the deliverables easier; in fact, expectations for clarity, completeness, proficiency, and professionalism are higher.

Analyze and summarize the following for EACH deliverable in your winter workshop **presentation**:

- Show the deliverable live – i.e. pull up your real Gantt chart, Decision Making Model, etc.
 - Can use snips in a PowerPoint for the presentations but the live version must be up and active so it can be referenced for the judges during Q&A
- Compare what you did last year to what you have for this. What is the same? Why did you keep it? What is different? Why did you change it?
- Evaluate the tools you used. Are they the same as last year? Why / why not?
- What went well? What didn't go well?
- Based on last year's performance, what did you change in PM to address any & all shortcomings?

Format: 1 hour Time Slot: You will walk through each deliverable pausing between each for judges questions. You can use up to 5 minutes to present your deliverable before the judges stop you for questions. You do not need to use the full 5 minutes for each.

- Intro- up to 5 minutes
- Gantt Chart -up to 5 minutes
 - Judges questions – 5 minutes
- WBS- up to 5 minutes
 - Judges questions 5 minutes
-Continued for other deliverables.....
- Summary – up to 5 minutes
 - Final comments/questions from the judges

Limited to 3 people presenting, the project manager (or the interim) must be present as one of those representatives. However, the full team is encouraged to come sit in the audience to support their team while presenting. Members from other teams cannot be present.

Final Competition

Teams submit a summary that explains what changed from winter workshop to the final for each deliverable and an overall summary on how the team thinks they did this year/will do at the final competition. Additionally, the team is asked to reflect on how they did across all 5 years of series two. This should not be a full report, only a summary with a maximum of 5 pages (not including a cover page or table of contents). No appendices. This will be due prior to the final competition to allow time for the judges to review prior to the presentation.

At the final competition teams present their deliverables to judges and Q&A session. The format of this session will be 8 minutes for the team to present and the remaining time for the judges to ask questions. Please keep in mind that the judges have read your report and summary, so the material that is presented should be supplemental to the report.

Limited to 3 people presenting, the project manager (or the interim) must be present as one of those representatives. However, the full team is encouraged to come sit in the audience to support their team while presenting. Members from other teams cannot be present.

Note: Understanding that AutoDrive series II is ending, it is foreseeable that the PM of a team might have already left the team for various reasons. If this occurs, the team must pick a replacement and have that person be the PM for the final presentation. If this occurs, it should be discussed in detail during the presentation time.

PM.1.3.2 Scoring Rubric

The judges will evaluate the project management deliverables based on the program Leadership scoring rubric found at <http://www.AutoDrivechallenge.com>.

Deliverable	Description	Total Point Value
Gantt Chart	Winter Workshop - 5min max presentation, 5 min max questions. - The PM should pull up the team's Gantt chart and walk through it live. - Detailed description of Gantt Chart created for year 5 that includes: - All AutoDrive milestones (each deliverable, each challenge, check-in points and final competition) - Bar graphs between the milestones with team member names attached to them, target and actual dates? - All interdependencies included (all tasks linked) - Is the critical path defined (shown in red) - Did they describe their detailed critical path analysis? - Did they answer the judges' questions sufficiently?	10
WBS	Winter Workshop – 5 min max presentation, 5 min max questions. - The PM should pull up the team's WBS and walk through it live. - Detailed description of their WBS for each team member including work completed prior to Winter workshop and projected work after. - Are all Gantt chart items tied to WBS elements (people or things)? - How are people currently or projected to be over/underutilized? How did you adjust for it? Provide examples of how you used the WBS. - Names and hrs/wk assigned to deliverables. - Did they answer the judges' questions sufficiently?	10
Team Structure and Communication	Winter Workshop – 5 min max presentation, 5 min max questions. - Pull up the Org Chart and walk through it. Then cover team communication. - Did they cover the organizational chart? - Does it include every team member? - Side chart showing any relevant supporting individuals - Academic advisors, technical advisors, lab technicians, etc. - Team roles are clearly defined with description - Project manager is specifically defined - Did they cover their communication model? - Defined ways of communication and description of how each were used - How did you communicate with people outside of your team? How do you plan to do this in the future? - Give information on how your change control process was implemented - A full meeting schedule for how work was accomplished is included	10

	○ Garage, Lab, and test track schedule is included • Did they answer the questions sufficiently?	
Decision-Making Model (DMM)	Winter Workshop - 5 min max presentation, 5 min max questions • The PM should pull up DMM model and talk to it. ○ Is a model included? ○ Can members reference it for all decisions? Does it include sub teams/multiple decision makers? ○ Does the decision model match the organizational chart or is it different? Is there an explanation as to why/why not? ○ Includes examples of how the DMM was used? • Did they answer the judges' questions sufficiently?	10
Summary	Winter Workshop – 5 min max presentation, 10 min max questions • Did they correctly summarize the entire project? ○ General overview of year 5 performance using PM terms ○ How is the team doing? How does the team think they will perform in year 5 final competition? What challenges does the team foresee for the remainder of the year? ○ What is going well this year? What isn't going well this year? • Did they answer the judges' questions sufficiently?	10
Final Competition Report	Maximum of 5 pages, not counting cover page or table of contents. Brief overview for each of the 4 deliverables • Explain any changes made over the year, especially any changes made between winter workshop and final competition. Or, if no changes were made, explain how the tools worked well (why were changes not needed). • Give an overall summary of how well each of the PM tools worked throughout this year. Gantt (2 points) WBS (2 points) Team Structure and Communication (2 points) DMM (2 points) Final Summary (4 Points) • How does the team think they did this year? How do you think the team will perform at final competition? • Total 5-year reflection, how satisfied is the team with their entire AutoDrive Series 2 performance?	12
Final Competition Oral Presentation and Q&A	The PM must be present. 8 min presentation covering: • Year 5 • The entire 5 years of AutoDrive Series II. Be reflective of how the team did year-by-year compared to expectations. Hint, the communication plan has directed the teams to archive each year.	13

	Use PM language and tools to show what was expected vs what is/was reality, is there anything you would have done differently.	
Total	PM Total	75

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Article D8: Systems Safety & Technical Reports

SSTR.1.1 Overview

The intent of this section is to familiarize the teams with what to expect during each year of the competition, from the perspective of safety work products deliverables.

During year 1, teams will have an opportunity to show a high-level understanding of their design, as well as initial safety analysis that will support their work towards building a safe and reliable autonomous vehicle. This year's work will form the foundation on which detailed safety work in following years will be built.

Once a design has been chosen, in year 2 teams can now conduct in-depth safety analysis to generate requirements. Requirements must clearly identify failures that could occur and provide a solution or mitigation strategy. Additionally, a traceability matrix will be used to track each requirement and the status of its validation.

For year 3, metrics must be defined to objectively describe the scenarios in which the vehicle is behaving "unsafe". Those metrics must be added to the requirements when applicable.

In year 4, students will focus on a report outlining the validation of previously defined safety requirements sourced from analysis techniques performed throughout the previous three years of competition. The validation status of requirements must be clearly reflected in the updated Requirements Traceability Matrix.

An additional focus of year 4 will be a report focusing on the identification of hazardous use cases from scenarios encountered in dynamic events and deriving any additional requirements from these scenarios within the Operational Design Domain. If the requirements derived from the use cases have already been identified through previous analysis, the source traceability in the requirement traceability matrix will be updated to reflect this. Any requirements outlined in the SOTIF Report should be traced and included in the Validation Report.

The final year of competition, year 5, is dedicated to summarizing all safety content (functional safety and SOTIF) into a final safety case report, outlining the argument for safety assurance and the achievement of safety goals through previous analysis, generation of requirements and validation. The safety case will be submitted and be peer reviewed by a panel of judges with feedback and questions sent to the teams. From the final safety case and in response to the feedback provided, teams will present their safety case to a panel of judges at final competition.

Phase	Deliverable	Point Allocation
Concept	Preliminary Hazard Analysis	100 points
Concept	Single Element Fault Analysis	75 points
Concept	Safety Concept	75 points
Requirements	Functional Interface Analysis	75 points
Requirements	Requirement Traceability Matrix and Validation Plan	100 points
Requirements	Safety Concept Update	25 points
Design	Hazard Metrics Translation	100 points
Design	Safety Concept Update	25 points
Design	Requirement Traceability Matrix and Validation Plan Update	75 points

Verification & Validation	Safety Validation Report	50 points
Verification & Validation	SOTIF Report	50 points
Verification & Validation	Requirement Traceability Matrix and Validation Plan Update	25 points
Final Safety Case	Safety Case Report	50 points
Final Safety Case	Final Safety Case Presentation	75 points

SSTR.1.2 Systems Safety & Technical Reports Year 1

SSTR.1.2.1 Preliminary Hazard Analysis

Preliminary Hazard Analysis or PHA, is the very first safety analysis that must be completed. This is a very useful tool because it defines your Safety Goals, and it does not need a detailed design to be completed.

To complete a PHA, each team must define what are the FUNCTIONS that their vehicle is expected to do. Teams must then analyze each function against HAZOP Guidewords. These guidewords will help the teams to define when a hazardous event is occurring. Each of those hazardous events must be assigned an Automotive Safety Integrity Level (ASIL) after analyzing their severity, exposure, and occurrence. Finally, safety goals and some requirements are defined for each hazard.

A diagram must be included to show the scope of the analysis.

An Example of PHA is shown in 1.7.1 as a reference only. Teams are encouraged to generate their own analysis.

SSTR.1.2.2 Single Element Fault Analysis

Once a system design has been drafted, Single Element Fault Analysis or SEFA can be performed. The goal of SEFA is to identify how failures in different parts of the design will impact functionality in other parts. The result of the analysis can then be used to define requirements or dictate changes in the design when needed. SEFA must include impact of each fault on the system, Resulting Safety Hazard, Mitigation Action, and Safety Requirement minimum.

A diagram must be included to show the scope of the analysis.

An example of SEFA is shown in Section 1.7.2, as a reference only. Teams are encouraged to generate their own analysis.

SSTR.1.2.3 System Safety Concept

System Safety Concept document is the holy grail of your safety work. It compiles the result of all your analysis, safety hazards, safety mitigations, restrictions, assumptions, etc. This document is continuously updated.

System Safety Concept must include the following content minimum:

1. Scope
2. System Description
3. Assumptions
4. Hazards and Safety Goals
5. Hardware Integrity
6. Software Integrity
7. Safe States

SSTR.1.3 Year 2

SSTR.1.3.1 Functional Interface Analysis

Functional Interface Analysis or FIA focuses on the interaction of different parts/systems of your design. The goal is to see if an error in each interface will result in a hazard (hazards that were defined in your PHA). If so, requirements must be added to diagnose or mitigate those hazards. Just like PHA, guide words are used here to assist with the analysis work.

A diagram must be included to show the scope of the analysis.

An example of FIA is shown in section 1.7.3, as a reference only. Teams are encouraged to generate their own analysis.

SSTR.1.3.2 Requirement Traceability Matrix and Validation Plan

Requirement Traceability Matrix or RTM compiles all your requirements from different analysis in one place. It also creates a link between your requirements and safety goals, and it provides a tracking system to see if those requirements have been validated.

It is important to note that a strong safety document does not have any stand-alone requirement. All requirements are generated to help mitigate the hazards associated with a safety goal. RTM can provide the evidence of the linkage between requirements and safety goals.

Requirement Traceability Matrix must include the following content minimum:

1. List of requirements and their IDs
2. Linkage of sub-requirements and requirements when applicable
3. Associated Safety Goal for each requirement
4. Validation plan for each requirement
5. Validation status of each requirement

An example of RTM is shown in section 1.7.4, as a reference only. Teams are encouraged to generate their own analysis.

SSTR.1.3.3 Safety Concept Update

The safety concept will be updated on an annual basis to ensure the documented safety strategy is up to date. Please ensure that updates and changes are captured in a change log within the document.

SSTR.1.4 Year 3

SSTR.1.4.1 Hazard Metrics Translation

Hazard Metrics Translation or HMT allows you to convert words into engineering values that your system and vehicle can be evaluated against. For example, during your PHA you might have defined a hazard as *Unintended Acceleration* and a subsequent safety goal of *System shall prevent Unintended Acceleration*. But what is Unintended Acceleration!

Here you can define that if vehicle acceleration is above ## m/s² for a longer than ## ms, then the vehicle has violated its unintended acceleration metrics.

HMT must be defined for all hazards defined in PHA that are rated above QM.

SSTR.1.4.2 Safety Concept Update

The safety concept will be updated on an annual basis to ensure the documented safety strategy is up to date. Please ensure that updates and changes are captured in a change log within the document.

SSTR.1.4.3 Requirement Traceability Matrix and Validation Plan Update

The requirement traceability matrix will be updated on an annual basis to ensure the any requirement additions/ changes are captured. Please ensure that updates and changes are captured in a change log within the document.

SSTR.1.5 Year 4

SSTR.1.5.1 Safety Validation Report

This report must show the tests that were conducted to validate each of the requirements from the requirement traceability matrix, the results of the tests, and the conclusions. Teams are encouraged to develop their own method/ template for documenting and sharing validation results.

Note: Validation can be done through physical validation or virtual simulation.

SSTR.1.5.2 SOTIF Report

Safety of the Intended Functionality (SOTIF) as defined in ISO 21448 aims to minimize unreasonable risk due to a hazard caused by functional insufficiencies. The SOTIF Report aims to drive additional safety requirements through analysis of the Operational Design Domain (ODD), applicable use cases and their minimal risk maneuver to meet the minimal risk condition.

Within this report, teams will:

- Outline the scope of their analysis used to develop the ODD, Use Cases and Minimal Risk Condition
- Define the ODD, outlining the operating conditions under which AV driving can be engaged.
 - Note: This is more than just blue light entry/ exit conditions, think about roadway types, weather conditions, etc.
- Define and detail applicable Use Cases for the ODD that are relevant to current and past dynamic challenges.
 - Include details such as vehicle speed, vulnerable road users, intersection, etc.
- Outline the minimal risk maneuver for the vehicle to achieve the minimal risk condition (stopped state) and define any additional requirements needed to successfully execute this maneuver.
 - If any additional requirements are sourced from this analysis, or if the requirements are common to those sourced through other analysis techniques, the requirement traceability matrix must be updated to reflect this.
- Define conclusions from the SOTIF Analysis and outline traceability from safety deliverables.
- Create a change log to track updates to the document.

SSTR.1.5.3 Requirement Traceability Matrix Update

The requirement traceability matrix will be updated on an annual basis to ensure the any requirement additions/ changes are captured. Please ensure that updates and changes are captured in a change log within the document.

SSTR.1.6 Year 5

SSTR.1.6.1 Final Safety Case

Complete a thorough report detailing the safety argument for your vehicle including both functional safety and SOTIF, this should reference your findings throughout all years of competition (from concept to validation results). In year 5, the teams will submit this document on 4/1/2026 for feedback from a Safety Review Board prior to final year competition.

SSTR.1.6.2 Final Safety Case Presentation

Following the submission of the Final Safety Case, a Safety Review Board will read and provide action items to the team prior to attending Final Competition. At Final Competition, teams will present their Final Safety Case to the review board with all action items addressed.

SSTR.1.7 Examples

Each year relevant templates will be shared with the teams for safety documentation.

SSTR.1.7.1 PHA

Functions: Provide Acceleration Control

HAZOP Guidewords:

1. **Loss of Activation (None when request):** Loss of Acceleration
2. **Incorrect Activation (More than Requested):** Excessive Acceleration
3. **Incorrect Activation (Less than Requested):** Insufficient Acceleration
4. **Incorrect Activation (Activation in Opposite Direction):** Not Applicable
5. **Unintended Activation (When none was requested):** Unintended Acceleration
6. **Locked Activation (Frozen Function):** Combination of above hazards

Hazard ID	Potential Vehicle Level Hazard	Hazard Description	Worst-Case Mishap Potential	Controllability or Other Mitigating Factors	Automotive Safety Integrity Level (ASIL) Assessment			System Safety Goal	System Safety Requirement
HAZARD_01	Unintended Acceleration	The vehicle accelerates at a rate that is greater than the rate expected by the operator. The unintended acceleration is requested by AV system erroneously.	Unintended vehicle acceleration may lead to collision with other vehicles, pedestrians, or objects in the vehicles path. At low to moderate speed the mishap potential may be high due to the closer proximity of other vehicles, pedestrians or objects, also the vehicle's capability to accelerate is typically greater at lower speed.	A greater distance between the subject vehicle and surrounding objects yields more time for the operator to control (mitigate) the situation. Braking systems with the capacity to overwhelm propulsion torque provide the operator with the means to slow or stop the vehicle. an emergency stop to terminate propulsion can assist the operator to control the situation Close proximity to other objects combined with high rates of unintended acceleration are more likely to panic the operator and reduce both the potential for proper control and the opportunity to use an escape path.	S3	E4	C3	ASIL D	System shall prevent Unintended Acceleration PHA_REQ_01: The AV system shall transition to safe state when unintended acceleration is detected PHA_REQ_02: No pedestrain, objects, or vehicles is permitted withing ##ft of the vehicle, when AV system is being activated. PHA_REQ_03:...

SSTR.1.7.2 SEFA

				PRIMARY Fault		RESULTING Fault															
Item Number	"Operating Scenario"	Function(s) (Responsibilities)	Element Operating States										Impact of Element Fault	Potential Safety Hazard(s)	Diagnostic Method(s) (If Applicable)	Mitigation Action(s)	SAFETY Requirements				
			Element name 1	Element name 2	Element name 3	Element name 4	Element name 5	Element name 6	Element name 7	Element name 8	Element name 9	Element name 10									
1	Element name 1	Element Functions	0	1	1	1	1	1	1	1	1	1									
2	Element name 2		1	0	1	1	0	1	1	1	0	1	1								
3	Element name 3				0																
4	Element name 4					0															

SSTR.1.7.3 FIA

List of suggested Guidewords:

Communication Related Guidewords
<i>Suggested Uses(s): Serial data interfaces</i>
Aged data (delay, repeated data, resequencing) Corrupted data (incorrect data, insertion of data, re-sequencing, masquerading) Loss of data Interrupted data (partial data transmit or no data transmit)
Electrical Related Guidewords
<i>Suggested Uses(s): Module-actuator, Module-sensors interfaces</i>
Open circuit Short circuit Out-of-Range (Low or High) Frozen/Stuck (Physically or Electrically)
Power Supply Related Guidewords
<i>Suggested Uses(s): Power Supply Interfaces</i>
Under/over voltage Loss of Power Drift Power spikes

Module	Interface	Direction	Interface Description	HAZOP Guide word	System Level Impact	Associated PHA Hazard	Hazard ASIL	Latent fault?	Safety Requirements	System Remedial Action	Comment
Sensor A	+12V and Ground	Roof PDB to Sensor A	Physical Power	Loss of Power	Reduced Perception	HAZ_13	ASIL B	No	FIA_REQ_01: Sensor A connector must have locking mechanism FIA_REQ_02: ...	Stop the Vehicle, Alert Driver, Use redundant sensor, etc.	
	Sensor State	Sensor A to Computing Module	CAN Communication	Aged data	Incorrect Object detection	HAZ_08	ASIL C				
				Loss of data	Reduced Perception	HAZ_08, HAZ_07	ASIL B				

SSTR.1.7.4 RTM

Requirement ID	Sub-Requirement ID 1	Sub-Requirement ID 2	Requirement Description	Safety Goal ID	ASIL	Validation Plan	Validation Status
PHA_REQ_01			Vehicle Shall transition to safe state when unintended acceleration is detected	SG_01	D	Vehicle Testing	Pending
	SEFA_REQ_08		Vehicle acceleration must be monitored and compared with requested value	SG_01	D	Bench Testing	Pending
		FIA_REQ_12	Loss of data from Acceleration sensor must be detected and driver must be alerted	SG_01	D	Bench Testing	Failed - TEST 12
	SEFA_REQ_09		Acceleration request must be computed redundantly and independently	SG_01	D	Design Analysis	Complete - System Design

Article D9: AutoDrive Challenge™ II Technical Inspection Testing & Verification for Years 3-5

Teams must receive 5 stickers indicating completion of all areas of Technical & Safety Inspections prior to being admitted onto the track for practice time or participation in Dynamic Events.

TI.1.1 Static Inspection Check List

Static Technical & Safety Inspections cover the following areas:

1. DVUL Level & Documentation
2. Occupant Safety
3. Driveability
4. Sensors & Visibility

Teams must complete all areas of Static Inspections prior to advancing to the Software Inspection. Detailed descriptions of inspection criteria can be downloaded and reviewed prior to competition.

TI.1.2 Software Inspection

Once a team has successfully received 4 stickers for Static Inspections, they are eligible to sign up for a software inspection on a closed course.

During Software Inspections, three people will be in the vehicle:

1. Safety Driver
2. One additional team member
3. An official Technical Inspector

Note: ALL drivers who will be participating in the competition must have taken the proper Driver Training.

Teams will be asked to enter autonomous mode and demonstrate that the Blue Autonomous Warning Lights activate automatically when autonomous mode is engaged. Then teams will be asked to disengage through takeover via both braking, acceleration and steering and demonstrate the vehicle is now in manual mode and the Blue Autonomous Warning lights automatically deactivate. Detailed descriptions of inspection criteria can be downloaded and reviewed prior to competition.

Note: After receiving the Software Inspection sticker teams may not change/ update code related to autonomous mode entrance/ exit conditions and blue light enablement. Violation of this will result in Tech & Safety stickers being revoked.

TI.1.3 Blue Autonomous Warning Light

Blue autonomous warning lights mounted to both the exterior and interior of the vehicle shall be used to indicate whether the vehicle is in autonomous mode or not. The lights shall be controlled via CAN messaging and shall have the ability to be off, solid blue, or blue flashing. The functionality of the lights will be checked during the Driving Test Inspection as described in section TI.1.2

TI.1.3.1 Exterior Blue Light

The exterior blue light shall be on the exterior of the vehicle, mounted to the roof. The exterior blue light serves as a warning to others around the vehicle that Autonomous Mode is active. **The light must be visible in daylight conditions by an observer on all sides of the vehicle and from at least 200 Feet away. The exterior light must also meet SAE J595 Class 1/2 or J845 Class 1/2 brightness specifications.** Teams should select their own mounting design strategy.

TI.1.3.2 Interior Blue Light

The interior blue light shall be on the interior of the vehicle, mounted on the dashboard. The interior blue light serves as an indicator to judges, passengers, and the safety driver that autonomous mode is active. **The light must be clearly visible in daylight conditions from any seat in the vehicle.**

TI.1.3.3 CAN Messaging

Teams must design into their autonomous control system the ability control the light using CAN messaging. The message structure and format will be provided in the scoring CAN dbc file. The lights must be connected to the scoring CAN bus. The Blue light controller must send a blue light alive signal via CAN. This signal will be used as an engagement condition in the AV state machine.

TI.1.3.4 Light Behavior

The control system shall turn on the interior and exterior blue light when Autonomous Mode is active and turn off the light when exiting Autonomous Mode for any reason, including when the Safety Driver performs a manual take over. The lights shall be solid blue (not flashing) when Autonomous Mode is active.

If the blue light(s) are receiving no communication from the compute platform, then they shall revert to a flashing blue light state indicating a loss of communication. If this state occurs, the loss of communication issue must be fixed before the vehicle is allowed to compete / practice in autonomous mode.

Blue light functionality will be tested during Tech Inspection and if it does not automatically function in this manner it will need to be resolved before passing Tech Inspection and continuing in the competition events.

Dynamic Event & Procedures

Article E1: Dynamic Challenges

DC.1.1 Overview

The goal of the Dynamic Challenges (DC) is for teams to demonstrate their ability to control their Chevrolet Bolt EUV and successfully navigate through basic highway and urban environments. Propulsion, braking, steering, and lighting control will be demonstrated along with the ability to integrate the perception control software developed in the Year 1-3 competitions. Teams will be required to utilize a compute platform, autonomous sensors, and interface equipment to the vehicle to complete these tasks. The challenge will be worth 500 points total and will consist of 4 parts: **V2X Challenge, Design Your Own Challenge, 99% Buy Off Ride Challenge, and the Localization Challenge.**

The **V2X Challenge** and **Design Your Own Challenge** will take place at your school's test facility in the fall and early wintertime and all other challenges will occur at the AutoDrive Challenge II competition in the summer. A scoring matrix and scoring CAN database will be posted separately from the rules in series resources. The scoring matrix will include a detailed list of the metrics which will be scored for each challenge and how much each of the metrics will be worth. The scoring CAN database will include the signals which will need to be sent on the scoring CAN bus during the challenges. Each challenge is defined below.

DC.1.1.1 Paddock Rules

- #1. The vehicle is never allowed to be in autonomous mode (no solid blue light) in the paddock. Not even for testing or trouble shooting.
- #2. The vehicle is never allowed to be in autonomous mode (no solid blue light) in the public areas. Not even for testing or trouble shooting.
- #3. The vehicle must have a wheel chocked at all times in the paddock, unless the vehicle has been approved to leave or enter the paddock by a paddock marshal. The preferred wheel is one that is easily observed by the paddock marshal standing by the paddock displays. Chocking more than 1 wheel is encouraged but not required.
- #4. No vehicle is allowed to move (at all) without a paddock marshal's approval and oversight.
- #5. Vehicle should quickly "beep-beep" the horn upon moving in the paddock or upon entering the paddock.
- #6. All occupants must have their seatbelts fastened before the vehicle moves, even in the paddock.
- #7. All windows should be down before and while driving in the paddock.
- #8. No running in the paddock.
- #9. No team members yelling in the paddock, unless it is an emergency.
- #10. Keep your area clean and tidy, pay attention to and minimize trip hazards.
- #11. Paddock marshal has final say on all paddock related questions.

DC.1.2 The Design Your Own Challenge

The **Design Your Own Challenge** will evaluate the teams' abilities to conduct vehicle testing and validation work using the basic AV control tasks that have been demonstrated during previous AutoDrive Challenge™ dynamic events. In this challenge you will be responsible for designing a test course at your school's AV test facility that combines various elements that you will be tested on in the June competition dynamic events. Your vehicle must run the course autonomously and then conduct an analysis of vehicles results from that test.

DC.1.2.1 Challenge Route

Two runs will be done for the DYOC this year.

Run 1: Vehicle will demonstrate the ability to navigate to multiple address points and recover quickly from a manual takeover.

The vehicle must enter AV mode and proceed autonomously to the 1st address point. After coming to a stop the 2nd address point must be input into the HMI and the vehicle should proceed to that address point. Prior to the 2nd address point a road blockage should be placed which forces the vehicle to come to a complete stop. After coming to a stop the safety driver should manually override and drive the vehicle manually to the 2nd address point. Then the 3rd address point should be input into the HMI and the vehicle should proceed autonomously to the 3rd (final) address point. The course may be designed as a loop if desired (so the third address point may be your starting point).

Run 2: Vehicle will demonstrate the ability to drive autonomously after a GNSS failure has occurred.

The vehicle must enter AV mode and proceed autonomously to the final address point. After proceeding for a fixed period (see scoring rubric) the GPS system should be attenuated in the manner you describe in your written report. The vehicle should then continue with an attenuated GPS to the destination or identify that it is not able to proceed and stop autonomously in a safe manner.

DC.1.2.2 Challenge Deliverables

Teams will be responsible for the following submissions:

- Pre challenge submission
 - A dimensioned drawing of your testing course using .kml or .kmz format which includes*:
 - GPS coordinates for position of all objects / obstacles
 - GPS coordinates of address points
 - GPS coordinates for planned route(s) / path(s)
 - GPS coordinates of lane lines
 - Change Requests
 - Changes are allowed post submission deadline, but changes must be approved by the rules committee (please submit a rules inquiry with the new .kml or .kmz file if you wish to make a change to your original course).

*Static objects should be included in the file as placemarks. Paths of dynamic objects and paths of the vehicle should be included as paths. All items should be clearly labeled

**The rules committee will review your design and give you approval to go forward with it. The rules committee reserves the right to require course changes to the course design.

- Challenge submission
 - A video recording the testing occurring.
 - Video must clearly show all dynamic obstacles, lane markings, signs, and traffic lights.
 - Video must show the interior and/or exterior lights blue lights functioning.
 - Video must show the HMI being used to input the destination address
 - CAN logs
 - Must include all vehicle and scoring CAN busses
 - Must be recorded using your NeoVI.
 - Written report*
 - Provide a writeup of how the team tested GNSS attenuation (1 page max)
 - How was attenuation achieved?
 - What were the effects?
 - How were the effects dealt with?
 - Provide a writeup of how your test data is analyzed (1 page max)
 - Describe what data you analyze and how you analyze it
 - Describe how you estimate the score you believe you should receive.
 - Give an estimated score based on the data you recorded using your NeoVI.
 - Provide a copy of the software that was used for doing the analysis

*This submission can use Jupyter notebook, SAE academic paper format, or other approved format (please submit a rules inquiry if you wish to use another format).

Event-specific submission instructions will be released prior to the due date.

DC.1.2.3 Course Design

The course must have the following characteristics:

- Overall length: ≥ 150 meters
- Address Points: ≥ 4
 - 4 locations along your route must be identified as address points that can be put into your HMI
 - One point must be the beginning of the route
 - One point must be the ending of the route
 - Two points must be between the beginning and end of the route
 - Each segment between address points must be a minimum of 30 meters in length and must have at least one turn.

- If you wish to do a loop, you may use the starting address point as the ending address point so long as the course meets all of the other requirements (in this case you would only have 3 address points).

*Additional address points may be included if you wish to have multiple ending / starting points or want to have destination addresses in the middle of the course

- Lane markings present: ≥ 50 meters
 - It is recommended that any portion of the course without lane markings present should have another method for visibly delineating the course boundaries (i.e. spray chalk, marker cones, etc)
- Curvature: ≥ 3 turns between 70° - 120°
 - 1 of the turns must be a left turn
 - 1 of the turns must be a right turn
- Intersection: ≥ 2
 - Both intersections must have a stop bar at the intersection entrance
 - The dimensions of the stop bar must be at least 2 m long and between 30 and 60 cm wide.
 - One intersection must have a traffic light
 - The light must at a minimum be a three can light (has green / yellow / red).
 - Lights with additional cans are permitted (i.e. doghouse, 4 can with protected left, etc)
 - Traffic light must cycle through green / yellow / red.
 - It is recommended the light be placed above the intersection, however side mounted light(s) are acceptable.
 - One intersection must have a stop or a yield sign
 - A traffic circle may be used as the stop / yield sign intersection
- Traffic barrel: ≥ 4
 - The end of the course must be fully blocked off by barrels
 - Barrels or barriers may be placed elsewhere along the course if desired
- Traffic sign: ≥ 3 (not including duplicates of the same sign type)
 - Sign types:
 - Right / Left Turn Only
 - Speed Limit
 - Teams may choose the speed limit value(s) for the speed limit sign(s)
 - Do Not Turn
 - Do Not Enter
 - Road Closed
- Pedestrian: ≥ 1
 - Pedestrian must start outside of the field of view and walk across the entire field of view perpendicular to the vehicle.
 - Teams may NOT use a live human test subject.

It is recommended that MMUTCD compliant testing tools are used for this challenge as that is what will be used during the June competition (i.e. signs, traffic signals, obstacles, etc). However, it is permissible to use items that are used in your geographic location (i.e. KPH speed limit signs, Canadian style traffic barrels, horizontally mounted traffic signals, etc).

The above requirements are the minimum requirements for the course. Innovation in testing is encouraged for this challenge.

DC.1.2.4 Blue Autonomous Warning Light

During this challenge whenever the vehicle is in autonomous mode the vehicle's Blue Autonomous Warning Lights must be illuminated. Policies regarding the blue autonomous warning light are available in section TI.1.2.

DC.1.2.5 Location

This challenge will be conducted remotely at your university test facility or another test facility of your choice (not Mcity).

DC.1.2.6 Challenge Timeline

Please reference DS.1.2 for deliverable due dates. You may submit change requests to the initial drawing at any point in time after the initial submission up until 3 weeks prior to the final Design Your Own Challenge final deliverable.

Submissions will be scored following the final due date using the last submission provided by the team. Late submissions due to weather will be subject to point deductions.

DC.1.3 100% Buy Off Ride Challenge

The **100% Buyoff Ride (BOR) challenge** will be a culmination of previous years' challenges, with an emphasis on real time perception and navigation that can handle dynamic situations. The goal of this challenge is to build upon previous dynamics events to demonstrate mature end-to-end autonomous capability loosely designed around simulating a ride sharing service. Additionally, vehicles will encounter a variety of environmental challenges and be required to autonomously navigate through those challenges.

DC.1.3.1 Route and Obstacles

In this challenge, teams will be given a route of several addresses (labelled points on the course) and will be required to successfully navigate to each address, while encountering a variety of dynamic and static obstacles and following all relevant traffic laws. At each address listed in a route, teams will be required to stop and brake hold until the next address point is selected. The vehicle must be stopped for a minimum of 5 seconds. Teams will be given one route to complete. Teams will be able to have a manual takeover and skip to the next address in the route at a point cost. Teams will also be given the option for a complete re-run but all points of the first run will be forfeited and only the second run will be scored, similar to previous years (see DCP.1.13).

A route, as used in this challenge:

- A route is a list of address points ex. A, D from figure below. The list of address points given in a route may not be contiguous.
- Teams must take the optimal route between two listed address points, where the optimal route is defined as the shortest mapped distance between two address points, accounting for any obstacles, road closures, etc.
- A list of all possible points will be provided at least 30 days prior to competition, but the scored route will be given one point at a time as the scored run progresses. As the vehicle reaches an address, the next address will be given and must be selected in the HMI. Teams must be prepared to navigate a route in real time without relying on collected map data during practice runs.
- A route may consist of up to 10 destination addresses and will be a minimum of 2 destination addresses.



Figure DC-1: Planned BOR Address Points

Please reference the Dynamic Challenge Address Points document in your series resources for official list of address points.

DC.1.3.2 Environmental Challenges

Along this year's BOR route the vehicles will be required to navigate through several environmental challenges that have not be encountered before.

Possible Environmental Challenges:

- Beach Boulevard – Roadway will have sand covering the pavement, obscuring the lane markings
- Obscura Drive – Roadway will be obscured by a heavy fog which may block the view of other vehicles, traffic control devices and the roadway itself.
- Sunset Ridge Road – A simulated sunset or sunrise will be occurring in this section of the roadway. The vision system on vehicles will need to deal with the glare from the sun in this section.

DC.1.3.2 Procedures, Definitions, and Figures

- Teams will be expected to stay in autonomous mode throughout the entire length of each route.
- Teams will encounter completely closed roads or road sections and must avoid construction to re-route and navigate to the next address point.
- Teams may not perform U-Turns.
- Teams will encounter traffic signals that will require your vehicle to deal with red, yellow, and green light configurations including flashing (red and yellow) and directional arrows.
- Teams will encounter both white and yellow lines, dashed and solid.
- Teams will encounter stop signs with and without limit lines and will be required to stop within a certain distance specified in the metrics.
- Teams will encounter both static and dynamic objects throughout the challenge and will have to react appropriately.
- Teams will encounter traffic control signs and will be required to react appropriately.
- Teams will encounter intersections both controlled and uncontrolled.

DC.1.3.3 Conclusion

The challenge is concluded when the vehicle reaches the final address point in the route and stops for a minimum of 5 seconds.

DC.1.4 Localization Challenge

DC.1.4.1 Looking Back

The Year 5 Localization Challenge will be extremely similar to the Year 4 Localization Challenge, a challenge that all teams struggled with. No team earned more than 15% of the available points. Therefore, this challenge has been brought into year 5 as a second attempt to demonstrate this ability.

DC.1.4.2 Background

The **Localization Challenge** will evaluate the vehicle's ability to operate in areas with limited GNSS (Global Navigation Satellite System) reception and therefore degraded GNSS performance. This is quite common in deep urban environments, canyons, tunnels, stacked roads, etc., proper Autonomous vehicle localization does not just rely on absolute localization from GNSS and IMUs, but also integrated relative localization from perception. This is a critical part of any Autonomous vehicle design and is especially critical to a Robo-Taxi's ODD (Operational Design Domain). One specific and extremely common environment that dramatically reduces the accuracy and general performance of any GNSS is the inner-city urban concrete canyons where Robo-Taxis typically operate. M-City has no ability to simulate this environment, therefore remotely activated electronic signal attenuators will be added to the GNSS antennas of each team's vehicle to emulate an inner-city concrete canyon GNSS degraded performance.

DC.1.4.3 Start

The Challenge will start off with zero attenuation of GNSS signals, GNSS working properly. Then shortly after the start of the challenge, scoring will remotely trigger the GNSS antenna attenuators to increase the attenuation of the GNSS signals, to simulate a **more than 10-meter** GNSS error. The teams are intended to continue through the challenge without this GNSS degradation impacting their performance.

The team's vehicle will be led to the starting line. At the starting line, a single destination waypoint will be provided. The teams are expected to select that waypoint from their HMI (note: a list of all possible waypoints will be provided in advance).

DC.1.4.4 Route and Obstacles

The vehicle should expect that the most optimum route could be blocked via a temporary road closure (barricade or barrels).

Vehicle may be required to interact with a balloon car for a period of time. Do not collide with the balloon vehicle! Collision results in a DNF.

DC.1.4.5 Conclusion

Once the vehicle has reached its waypoint destination, the vehicle should automatically apply the parking brake and turn on the hazard lights to signal the run is over. Points will be awarded for applying the parking brake and points will be awarded for turning on the hazards.

Note: if at any time during the challenge, the vehicle cannot go any further, applying the parking brake and turning on the hazards to signal the end of the run (before leaving autonomous mode) will receive the same points that applying the parking brake and turning on the hazards would receive at the final destination.

DC.1.4.6 General

Points will be awarded once each decision point (intersections), obstacles, or traffic control devices, has been passed (cleared). See Scoring Rubric.

Scoring triggered GNSS attenuation will be available during June Competition practice times.

For testing at the team's home track, there are many different ways of attenuating GNSS antennas, from inline L band attenuators to much more simple and crude methods such as disconnecting the antennas or placing physical objects over the GNSS antennas – i.e. covering the antenna with cooking pots attached to hockey sticks. It is recommended that teams practice with GNSS degradation.

DC.1.5 HMI Challenge

DC.1.5.1 Overview

In Year 5 teams will be required to design and use a human-machine-interface (HMI) system for enabling and disabling their autonomous systems which utilize physical, or touchscreen button presses on a purpose-built interface. Teams will also be required to use their HMI systems to enter route addresses/destinations for the Localization and 100% Buy-off Ride Challenges. Prior to the challenges teams will be provided with addresses in the form of GPS coordinates which must be selectable at the starting line from their HMI.

Using keyboards / mice to run terminal commands or to start up processes manually is not considered a discrete button input. All controls must be accessible to the safety driver.

DC.1.5.2 HMI Restrictions

Teams may use no more than 5 interactions with their HMI at each stop, including the starting line.

Examples of acceptable methods of entry include but are not limited to: virtual buttons on a touch screen display, physical buttons that interact with the HMI display, rotary encoder that interacts with the HMI display. Scrolling or swiping through a touchscreen or turning a rotary encoder do not count as an interaction. Interactions are when the button is selected.

If you have any questions whether your input method is acceptable, please submit a rules inquiry.

DC.1.5.3 HMI Visualization

For Year 5 teams are required to create a visualizer screen for their HMI that displays the planned route and surrounding obstacles in the HMI.

The planned route should be overlaid on top of a background map of the entirety of MCity. Options for the background map include utilizing the DMP HD Map, Open Street Map, Satellite View, etc.

The instructions for the planned route should be updated onto the screen. The upcoming maneuver must be clearly visible, and the following steps must be viewable with a button press that "collapses" the route plan.

Obstacles should be identified in the visualization as well. These obstacles include: animal, pedestrian, construction, railroad, traffic light (with color), stop signs, and detected speed limit.

DC.1.6 V2X

There will be no standalone V2X challenge this year, points will be awarded in the BOR / Localization challenges for successfully detecting traffic lights via V2X. Teams are encouraged to utilize V2X capabilities in their AV to help navigating both challenges.

Article E2: Dynamic Challenge Procedures

DCP.1.1 Vehicle

All teams will be provided Chevrolet Bolt EUV along with a complete document package related to sub systems of the vehicle.

Through the <http://www.AutoDriveChallenge.com> General Motors will provide materials relevant to team's participation for AutoDrive Challenge™ II with the understanding that these are to be used for the competition only. Given that this is intellectual property downloads will be limited to team members, faculty advisors, graduate assistant, and team captains. As per the NDA signed by each member in these categories you are not permitted to distribute this information for any other use than AutoDrive Challenge™ II.

Teams will receive the following components in their package:

1. CAD drawings of exterior vehicle surfaces and selected interior vehicle surfaces
2. Electrical and wiring schematics
3. Vehicle physical and miscellaneous properties (mass, center of gravity, tire stiffness etc.)
4. DBC Files containing necessary Tx/Rx messages to interact with steering, braking, acceleration, and lighting control modules.
5. An instructional guide to interfacing with vehicle serial data:
 - a. Higher level data flow diagrams
 - b. Low level specifics on using the individual data signals
6. Vehicle information documents
 - a. Owner's Manual
 - b. MyLink Infotainment System Guide
 - c. Maintenance Schedule
 - d. Get To Know guide

Teams must design to allow for 3 persons to fit in the vehicle:

- A safety driver,
- 1 other team passenger

- An in car judge during the scored runs.
- Build envelope is one foot off the bumpers and one foot past the side view mirrors.
- Space for a judging DAC will be marked and cannot be encroached on.

During the practice runs teams may have up to 4 team members in vehicle space permitting. During the scored runs teams may have up to 3 team members in vehicle space permitting. The middle rear seat will be used for data acquisition and may not have anyone sitting in it during any practice or scored runs. Seat belts must be worn at all times while the vehicle is not wheel chocked.

DCP.1.2 Sensors

Each team will utilize a variety of sensors for the dynamic challenges. Details on any sponsored sensors will be provided in the series resources. There are no requirements around what sensors or sensor technology can be used for each challenge outside of any sponsorship related restrictions (you are free to choose the sensor technology and suite you think would best accomplish the goals of the challenges).

The sensor suite you design must be able to be used for all the challenges. Sensors may not be added, removed, or moved for any individual challenge, but every sensor present on the vehicle does not need to be used for every challenge.

The teams GNSS receiver must use a SMA or TNC connector on the receiver side of the GNSS antenna connection(s). If you cannot use one of these connector types please submit a rules inquiry.

DCP.1.3 Compute Platform / Electronics

Intel will be providing the compute platform for AutoDrive Challenge II. Teams are responsible for integrating the compute platform into their vehicles for the challenges. CAN messaging will be used for sending autonomous driving commands to the competition vehicle and for transmitting processed perception data to the scoring system for scoring at the event.

DCP.1.3.1 CAN Interface

The compute platform will need to have the ability to transmit on 3 high speed (dual wire) CAN busses and 1 low speed (single wire) CAN bus. Two of the high-speed CAN busses and the low-speed CAN bus will be used to send autonomous driving commands to the Bolt EUV (the HS, CE, and LS bus respectively). The other the high speed can bus will be used to send scoring signals. The scoring bus must be separate from the CAN busses used to control the vehicle. However, it may be a shared bus with any CAN based sensors or equipment that is used. Information on the CAN message structure and hardware interface will be provided in the series resources on www.AutoDrivechallenge.com.

DCP.1.3.2 Power Consumption

There will be a 12 VDC power connection provided in the trunk of the vehicle connected to the vehicle battery and auxiliary power module (APM). This connection will have a 60-amp fuse which will allow for a power consumption of ~720 Watts. The 60-amp fuse may be replaced by an equivalent model 100-amp fuse if additional power is required.

Depending on the ambient conditions (during extreme heat or cold) and vehicle settings (climate control / heated seats / heated steering wheel usage) the APM may not be able to provide 1,200 Watts continuously. If the 12-volt car battery is being drained during testing power consumption should be reduced or an external battery power supply / storage device may be added by the team.

DCP.1.3.3 Cooling

The competition will take place during June in Michigan and there may be high outside temperatures. Care should be taken to ensure the compute platform and other electronics can stay sufficiently cool during the competition runs. Delays in scored runs and practices will not be given for teams that require a cooldown time.

DCP.1.3.4 EMI

Good engineering practices should be used when designing and building the electrical system and associated wiring. Care should be taken to avoid excessive EMI from your system, especially EMI which interferes with Wi-Fi, GPS, and CAN communications. If EMI from your vehicle interferes with the scoring or other competition systems, you will be required to fix or remove the source of the EMI interference before your team will be allowed to compete.

DCP.1.4 Starting Order

Teams will draw for their starting order at the Winter Workshop, and it will be posted in the Student Handbook prior to competition. The organizing committee will determine the allowable windows for each run and retains the right to adjust for weather or technical delays. Teams who have not run by the close of the event will receive 0 points for the Challenge unless the delay is due to weather or other approved causes.

DCP.1.5 Starting

A verbal indication from the in-car judge will be used to indicate the approval to begin. Once approval to begin is given, the teams are to activate their vehicle systems. Timing begins once you are given the verbal indication from the in-car judge. Once you arrive at the starting line no further setup or testing can be done (i.e. you cannot move your vehicle around to adjust your GPS at the starting line).

DCP.1.6 Software and Calibration Changes

During the Dynamic Event days teams are permitted to make software and calibration changes except during scored runs.

DCP.1.7 Peripheral Input Devices

DCP.1.7.1 Peripheral input devices after leaving the pits

After leaving the pit area teams will not be allowed to utilize peripheral input devices other than what is described in the HMI section. If a different peripheral input device is used a point penalty will be given (see scoring rubric). The time at the starting line will be limited so teams are encouraged to design a system that can be enabled and enter autonomous mode easily.

DCP.1.7.2 Peripheral input devices after crossing the start line

Once the vehicle has crossed the starting line during a dynamic event **NO** compute platform peripheral input devices may be used or accessed until the dynamic event is completed or the team has ended their run. No exceptions will be made for data acquisition / viewing or any other reasons (the AV system should be designed to operate autonomously).

Peripheral input devices include but are not limited to: mouse, keyboard, touch screen, joystick, game controller, analog switches, etc.

Violation of this rule will result in a **score of 0** for the run in which the violation occurs.

DCP.1.8 Remote Support

No remote support is allowed during scored runs once runtime begins. This includes (but is not limited to) remote connection to the servers, phone calls, instant messaging services (text, MS Teams, Slack, etc.).

DCP.1.9 Vehicle Integrity and Disqualification

During the Dynamic Events, the mechanical integrity of the vehicle must be maintained during all challenges. Any vehicle condition that could compromise vehicle integrity or could compromise the track safety or pose a potential hazard to participants, spectators, and volunteers will be a valid reason for exclusion by the official until the problem is rectified.

Vehicles and all AV equipment added to the vehicle must be designed to survive vibrations and high-g dynamic maneuvers (~1 g) during the calibration procedure and during vehicle transportation to and from the competition site. Everything in the vehicle must be secured (backpacks, fire extinguisher, keyboards, monitors, etc.). Vehicles may be driven in circles, figure eights, or other maneuvers during the calibration procedure.

DCP.1.10 Access to Dynamic Event Areas

The organizing committee will specify areas of the event as dynamic event areas or as waiting areas for dynamic events where only team members wearing official dynamic passes may gain access and the number of tools that may be used in this area may be restricted. Please refer to the student handbook to understand the specific restrictions on this competition.

Persons found in violation any written or verbal dynamic event access rules (entering a closed course, entering a dynamic area without a pass, entering a dynamic area closed to your team, etc.) will subject your team to a 25-point penalty per violation. Team leaders are responsible for ensuring that all team members on site are aware of the dynamic area rules including rules mentioned during the captain's meetings.

DCP.1.11 Weather Conditions

The organizing committee reserves the right to alter the conduct and scoring of the competition based on weather conditions to maintain safety of the participants and volunteers at any given time.

The vehicle should be designed to operate in light rain in the event of inclement weather during the competition.

DCP.1.12 Captain's Meeting

A captain's meeting will occur on every practice / scored run day at the event. Your team captain or a designee is required to attend the captain's meeting to be eligible to compete in that day's dynamic events / practice. If you are unable to attend the captains meeting, a makeup meeting must be attended with the dynamic captain before you will be permitted to enter the course.

DCP.1.13 Did Not Finish (DNF) Types and Impacts to Challenges

Unless otherwise stated in the scoring rubric or elsewhere in the rules all of the DNFs apply to all dynamic challenges.

DCP.1.13.1 Autonomy Computer Interaction

Occurs when a team member interacts with any vehicle autonomy control system after crossing the start line during a scored challenge attempt.

DCP.1.13.2 Off course

Occurs when the vehicle has all four (4) tires outside the course boundary as indicated by solid white lines, yellow lines, or pavement edge.

DCP.1.13.3 Collision

Occurs when the vehicle collides with any object or drives on top of a curb.

DCP.1.13.4 Exceeding Mcity Speed Limit

Occurs when the vehicle exceeds the Mcity site speed limit (25mph)

DCP.1.13.5 Exceeding the Challenge Time Limits

Teams will have a maximum of 20 minutes to complete the scored challenges. Timing begins per DCP.1.5. The Challenge timer will stop once the vehicle reaches the Finish line of the final segment of the route within the challenge. Any points earned up to that point are kept. Exceeding the challenge time limit ends your challenge attempt for that day.

Teams will have a maximum of 10 minutes to cross the starting line of the challenge in AV mode. Timing begins per DCP.1.5. Exceeding the challenge start time limit ends your challenge attempt for that day.

DCP.1.13.6 Manual Takeover

Occurs as a result of any manual takeover or E-stop activation while on-course.

DCP.1.13.7 Vehicle Integrity and Disqualification

During the Dynamic Events, the mechanical & technical integrity of the vehicle must be maintained during all challenges. Any vehicle condition that could compromise vehicle integrity or could compromise the track safety or pose a potential hazard to participants, spectators, and volunteers will be a valid reason for exclusion by the official until the problem is rectified.

If the vehicle at any time displays behavior exhibiting non-compliance of the technical inspection items, the tech inspection stickers may be removed at the judge's discretion and the vehicle will no longer be allowed to compete unless the vehicle is reinspected and demonstrates a return to compliance with the original inspection.

DCP.1.14 Reruns

Teams will have the option to complete one additional attempt within the original 20-minute time window for each challenge. Scores for the initial attempt will not be provided, and a re-run attempt will void any possible score received from the initial attempt at the time when the vehicle crosses the start line for the re-run attempt. The 20-minute time window will not be paused during the re-run attempt staging. The original 20 minute time window will not be extended unless there is a delay due to scoring or course setup.

The event captain may disallow a re-run if the vehicle is deemed unsafe due to first-run performance, collision, revocation of tech inspection sticker, or safety requirement violations.

Article E3: Scoring & Protests

SP.1.1 Official Scores

All official AutoDrive Challenge II scores will be posted on the official webpage, and on the AutoDrive Challenge II App.

SP.1.2 Problem Reports

It is recognized that thousands of hours of work have gone into all aspects of the competition and teams are entitled to all the points they can earn. We also recognize there can be differences in the interpretation of rules, the application of penalties, and the understanding of procedures among volunteers and participants. The Organizing Committee will make every effort to fully review all questions and resolve problems and discrepancies.

It is also recognized that the measuring and evaluation of the many metrics of an AutoDrive Challenge II team's performance while executing dynamic events is a very complicated process. Errors can also occur during the tabulation and processing of scores for static events. The Organizing Committee has worked diligently to minimize the risk of mishaps during the collection of results data and the rendering of scores; however, the possibility of a data or procedural error certainly does still exist.

After scores or results have been posted, if a team believes a non-trivial error may have occurred during the processing of their team's results which has had a substantive effect on their score, a member of that team is permitted (and encouraged) to file a scoring "Problem Report" describing the situation. The Problem Report officially notifies the Organizing Committee of a concern related to one or more results and provides the details necessary for the appropriate personnel to begin a review of the results and scores associated with the problem described.

Problem Reports may be filed within 24 hours of the posting of the affected result or score. After 24 hours have passed from the initial posting of the result or score Problem Reports are no longer accepted. To file a Problem Report, please visit www.AutoDriveChallenge.com/go/pr

SP.1.4 Decisions

The decision of the competition Organizing Committee regarding any protest is final.

Article E4: Vehicle Numbers & Livery

The organizers and officials use vehicle numbers to positively identify team vehicles. Teams must design numbers to be visible in all race conditions or keep them clean and conspicuous. Numbers shall not be obscured by any other portion of the vehicle.

VN.1.1 Required Numbers

Two primary numbers are required to be securely affixed to the car. The vehicle's number shall be readily visible from the left side and right side of the vehicle and strongly contrast with the number's background color.

VN.1.2 Required Font

Vehicle numbers shall be displayed in either the "Highway Gothic Regular" font or "Century Gothic Bold" font. No other fonts are permitted. Examples of both fonts are given below.

Highway Gothic: **1 2 3 4 5 6 7 8 9 0**

Century Gothic Bold: **1 2 3 4 5 6 7 8 9 0**

Figure VN-1: Approved vehicle number font examples.

VN.1.3 Location

VN.1.3.1 Side Numbers

Side numbers, mounted to the left and the right sides of the vehicle, shall be mounted on the passenger and driver side door. Side numbers shall not be visually obstructed by any part of the vehicle.

VN.1.3.2 Front / Rear Number

Front / rear numbers may be added to the vehicle but are not required.

VN.1.3.4 Orientation

Numerals shall be aligned along a common horizontal line, and the entire number shall be mounted in a generally horizontal orientation to facilitate fast vehicle identification. The number shall have 25.4 mm (1.0 in.) spacing between numerals.

VN.1.3.5 Number Size

The numbers must be at least 152 mm (6.0 in) high, dimension "H" in Figure X. The numbers shall have a stroke width in proper proportion to the font design for the given character height.

VN.1.3.6 Background

Each number on the vehicle shall have a highly contrasting background to facilitate easy reading. The edges of the backing panel shall be no less than 25.4 mm (1.0 in.) from the edge of the numbers (dimension "D" in Figure X). Numbers may be outlined to provide enhanced contrast.

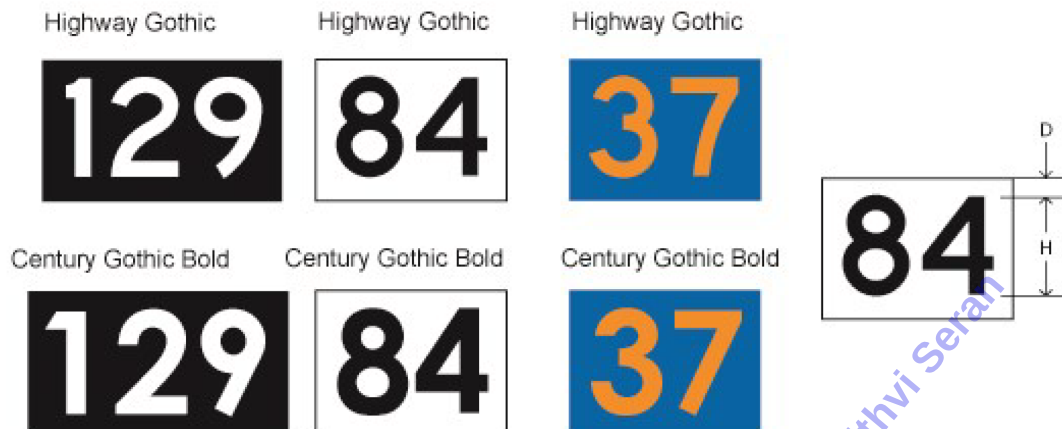


Figure X: Vehicle Numbers, Example

VN.1.3.7 Number Color

Teams are free to select their own number and backing colors, provided that the colors are high contrast and facilitate fast vehicle identification. The number backing shall be all one color and all numerals shall be of matching color. See Figure X.

VN.1.4 SAE International & General Motors Logos

Two (2) SAE International & General Motors logos must be displayed on the vehicle in prominent locations. These will be distributed during registration at the competition.

VN.1.5 Sponsor Identification

Teams may display advertising from their vehicle's sponsors, provided it is in good taste and does not conflict with the vehicle's number. SAE International will require all entrants to display advertising from the competition sponsors, which will be made available to teams.

VN.1.6 Vehicle Livery

Teams may display a livery / vehicle wrap to personalize your vehicle to your school / team, provided it is in good taste and does not conflict with the vehicle's number or sponsor logos.

APPENDIX

SAE Standards Available to Teams

AutoDrive Challenge™ II

J3016 – Taxonomy and Definitions for Terms Related to On-Road Motor Vehicle Automated Driving Systems

J3018 - Guidelines for Safe On-Road Testing of SAE Level 3, 4, and 5 Prototype Automated Driving Systems (ADS)

J3063 - Active Safety Systems Terms & Definitions

Baja SAE

J586 - Stop Lamps for Use on Motor Vehicles Less Than 2032 mm in Overall Width

J759 - Lighting Identification Code

J994 - Alarm - Backup – Electric Laboratory Tests

J1741 - Discriminating Back-Up Alarm Standard

Clean Snowmobile Challenge

J192 - Maximum Exterior Sound Level for Snowmobiles

J1161 - Sound Measurement – Off-Road Self-Propelled Work Machines Operator-Work Cycle

Formula SAE Hybrid

J1318 - Gaseous Discharge Warning Lamp for Authorized Emergency, Maintenance and Service Vehicles

J1673 - High Voltage Automotive Wiring Assembly Design

Formula SAE

SAE 4130 steel is referenced but no specific standard is identified

SAE Grade 5 bolts are required but no specific standard is identified

Supermileage

J586 - Stop Lamps for Use on Motor Vehicles Less Than 2032 mm in Overall Width

SAE Technical Standards for Supplemental Use

Standards Relevant to Baja SAE

J98 – Personal Protection for General Purpose Industrial Machines – Standard

J183 – Engine Oil Performance and Engine Service Classification - Standard

J306 – Automotive Gear Lubricant Viscosity Classification - Standard

J429 – Mechanical and Material Requirements for Externally Threaded Fasteners – Standard

J512 – Automotive Tube Fittings - Standard

J517 – Hydraulic Hose - Standard

J1166 – Sound Measurement – Off-Road Self-Propelled Work Machines Operator-Work Cycle

J1194 – Rollover Protective Structures (ROPS) for Wheeled Agricultural Tractors

J1362 – Graphical Symbols for Operator Controls and Displays on Off-Road Self-Propelled Work Machines - Standard

J1614 – Wiring Distribution Systems for Construction, Agricultural and Off-Road Work Machines

J1703 - Motor Vehicle Brake Fluid - Standard

J2030 – Heavy Duty Electrical Connector Performance Standard

J2402 – Road Vehicles – Symbols for Controls, Indicators and Tell-Tales – Standard

Standards Relevant to Clean Snowmobile Challenge

J44 – Service Brake System Performance Requirements – Snowmobiles - Recommended Practice

J45 – Brake System Test Procedure – Snowmobiles – Recommended Practice

J68 – Tests for Snowmobile Switching Devices and Components - Recommended Practice
J89 – Dynamic Cushioning Performance Criteria for Snowmobile Seats - Recommended Practice
J92 – Snowmobile Throttle Control Systems – Recommended Practice
J192 – Maximum Exterior Sound Level for Snowmobiles - Recommended Practice
J288 – Snowmobile Fuel Tanks - Recommended Practice
J1161 – Operational Sound Level Measurement Procedure for Snowmobiles - Recommended Practice
J1222 – Speed Control Assurance for Snowmobiles - Recommended Practice
J1279 – Snowmobile Drive Mechanisms - Recommended Practice
J1282 – Snowmobile Brake Control Systems - Recommended Practice
J2567 – Measurement of Exhaust Sound Levels of Stationary Snowmobiles - Recommended Practice

Standards Relevant to Formula SAE

J183 – Engine Oil Performance and Engine Service Classification - Standard
J306 – Automotive Gear Lubricant Viscosity Classification - Standard
J429 – Mechanical and Material Requirements for Externally Threaded Fasteners – Standard
J452 - General Information – Chemical Compositions, Mechanical and Physical Properties of SAE Aluminum Casting Alloys – Information Report
J512 – Automotive Tube Fittings - Standard
J517 – Hydraulic Hose - Standard
J637 – Automotive V-Belt Drives – Recommended Practice
J829 – Fuel Tank Filler Cap and Cap Retainer
J1153 - Hydraulic Cylinders for Motor Vehicle Brakes – Test Procedure
J1154 – Hydraulic Master Cylinders for Motor Vehicle Brakes - Performance Requirements - Standard
J1703 - Motor Vehicle Brake Fluid - Standard
J2045 – Performance Requirements for Fuel System Tubing Assemblies - Standard
J2053 – Brake Master Cylinder Plastic Reservoir Assembly for Road Vehicles – Standard

Standard Relevant to Formula Hybrid

J1772 – SAE Electric Vehicle and Plug in Hybrid Conductive Charge Coupler

Standard Relevant to all CDS Competitions

J1739 – Potential Failure Mode and Effects Analysis in Design (Design FMEA) Potential Failure Mode and Effects Analysis in Manufacturing and Assembly Processes (Process FMEA) and Potential Failure Mode and Effects Analysis for Machinery (Machinery FMEA)